

3

Lenses

In this chapter, we will learn about

- the image formation of convex lenses
- the image formation of concave lenses
- the properties of the image formed by lenses
- Ext the lens formula

3.1 Convex and concave lenses

Do you know how a magnifying glass works? It contains a piece of **curved** glass object called **lens** which can make objects look bigger (Fig 3.1a). A drop of water also behaves like a lens (Fig 3.1b). Their magnifying effect is due to the refraction of light.



Fig 3.1a *The flower looks bigger when viewed through a magnifying glass.*



Fig 3.1b *The surface of a leave is enlarged when viewed through drops of water.*

Apart from the magnifying glass, lenses are used in many optical instruments that we use every day (Fig 3.1c).

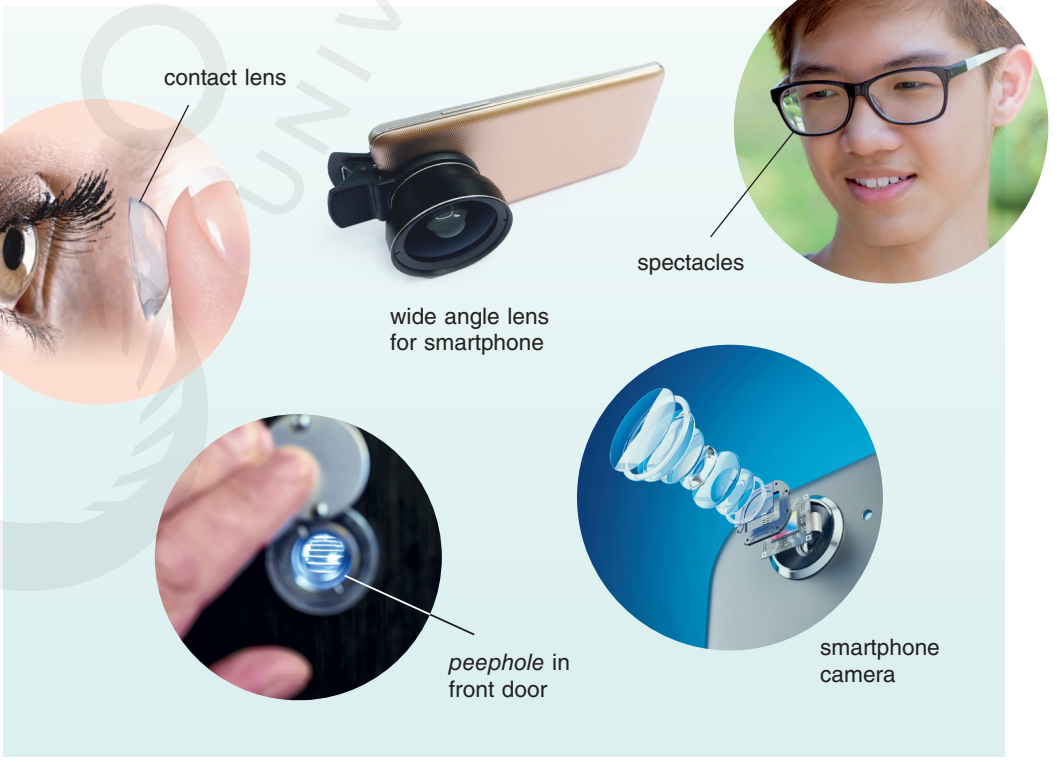


Fig 3.1c *Some optical instruments that contain lenses.*

lens 透鏡　　peephole 防盜眼

1 Bending of light in lenses

It may help you to remember 'convex' by the V-shape edge of the lens and 'concave' by its 'cave shape'.

Lens can be classified into two types. A **convex lens** is thicker in the middle (Fig 3.1d); a **concave lens** is thinner in the middle (Fig 3.1e).

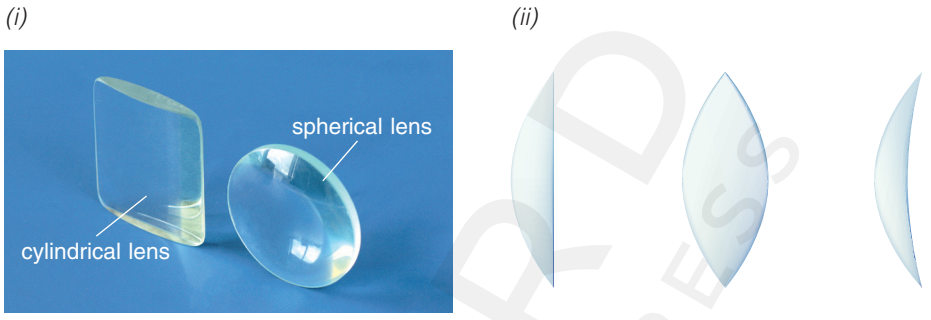


Fig 3.1d *(i) Photos and (ii) top views of some convex lenses.*

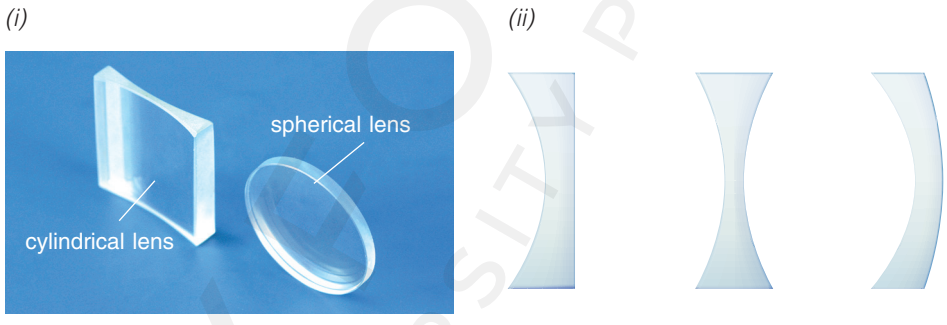


Fig 3.1e *(i) Photos and (ii) top views of some concave lenses.*

Experiment 3a Bending of light in a convex lens and a concave lens

Video

→ This video demonstrates Expt 3a.

1 Set up the apparatus as shown (Fig a). Direct parallel light rays onto a convex lens at different angles. Observe how the rays are bent.

2 Direct a light ray onto different positions of the convex lens (Fig b). Observe how the ray is bent.

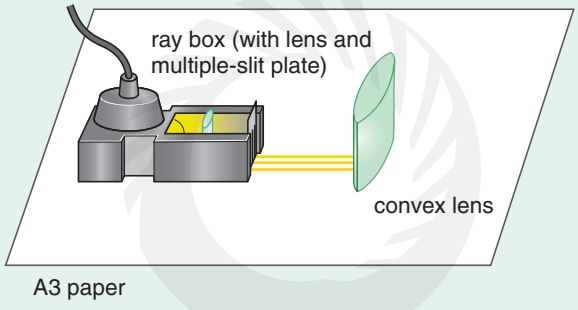


Fig a

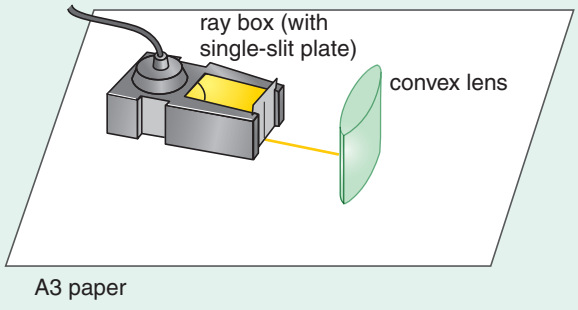


Fig b

Do not touch the hot ray-box lamp.

3 Repeat steps 1 and 2 with a concave lens.

Discussion

How do convex and concave lenses bend light rays?

convex lens 凸透鏡 concave lens 凹透鏡

Caution

When rays converge to a point after passing through a convex lens, they do not stop there but continue to travel straight on.

A convex lens converges **parallel rays** to a point, no matter which direction the rays are directed from. Therefore, a convex lens is also called a **converging lens**. Figure 3.1f shows how the parallel light rays converge. For simplicity, we use a symbol to represent the convex lens.



Fig 3.1f *A convex lens converges parallel rays.*

On the other hand, a concave lens diverges **parallel rays**, so it is also called a **diverging lens**. When the emerging rays are extended backwards, they appear to come from a point (Fig 3.1g). Again, we use a symbol to represent the concave lens.

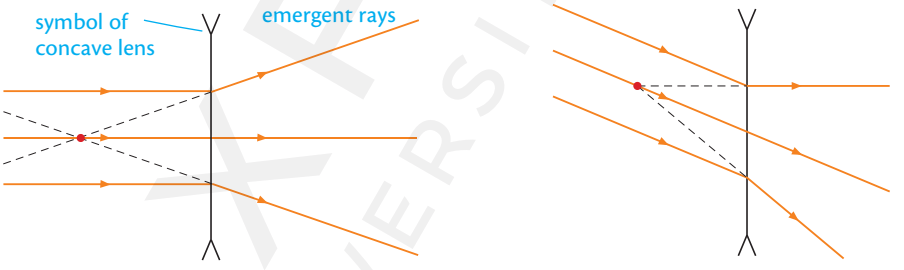
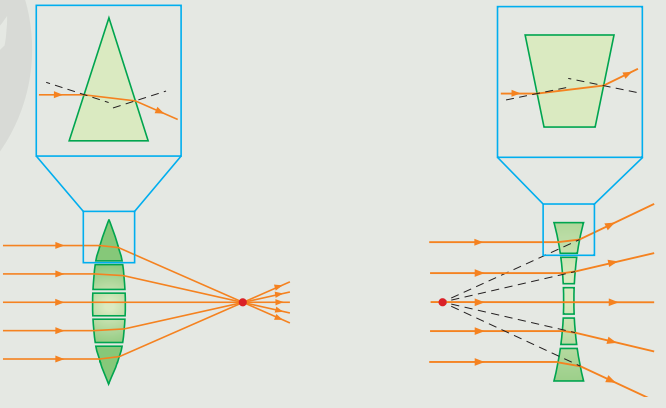


Fig 3.1g *A concave lens diverges parallel rays.*

Know more

Why a lens bends light rays

We can think of a lens as a series of prisms. Refraction occurs as light enters and leaves the prism. Note that the laws of refraction are applicable to the bending of rays for both convex and concave lenses.



Ts may ask Ss how to draw the normal to a curved surface.

✎ After introducing the principal axis, Ts may emphasize that a convex lens bends a light ray towards the principal axis while a concave lens bends a light ray away from the principal axis.

A light ray changes direction after passing through a lens except when it passes through the central position of the lens (Fig 3.1h). This position is the **optical centre** *O* of the lens.

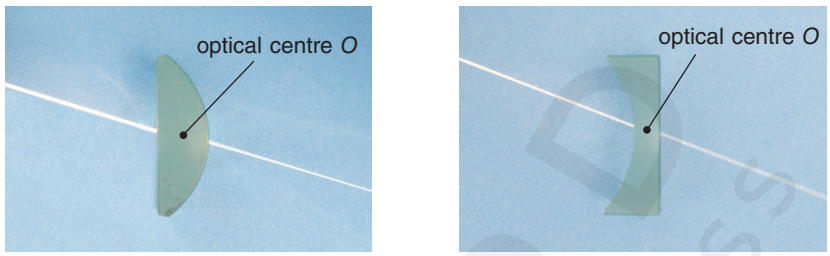


Fig 3.1h *A ray passing through the optical centre does not change its direction.*

Other features of a lens include the **principal axis**, the **principal focus**, the **focal plane** and the **focal length** (Fig 3.1i).

✎ Remind Ss that the rays meet and cross over at the focus *F* after passing through a convex lens. Some Ss may think that the rays converge to and 'stop' at *F*.

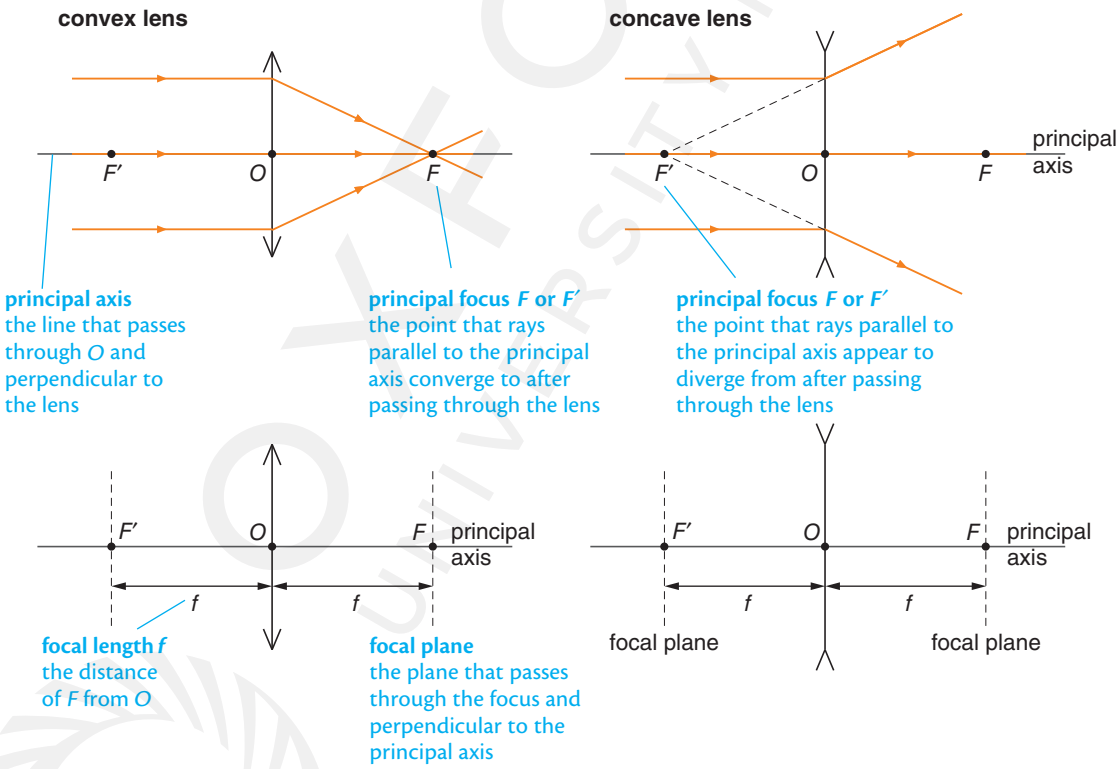


Fig 3.1i *Key features of lenses.*

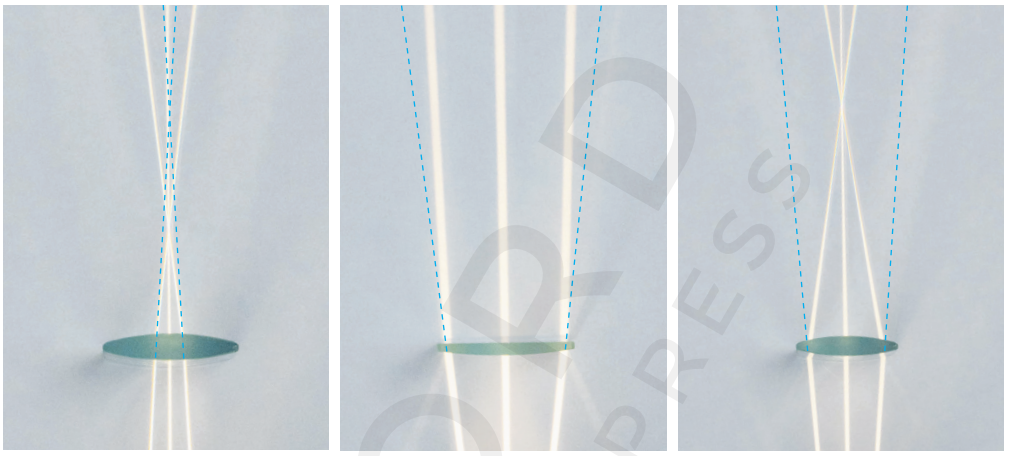
Note the following:

Foci is the plural form of focus. ▶

1. Each lens has two principal foci *F* and *F′*, one on each side of the lens. The foci are equidistant from *O*.
2. A lens with a larger curvature bends light more and therefore has a shorter focal length.
3. A lens with a greater refractive index *n* bends light more and therefore has a shorter focal length.

optical centre 光心　principal axis 主軸　principal focus 主焦點
focal plane 焦平面　focal length 焦距

In general, a convex lens bends light rays towards the principal axis (Fig 3.1j) while a concave lens bends light rays away from the principal axis (Fig 3.1k).

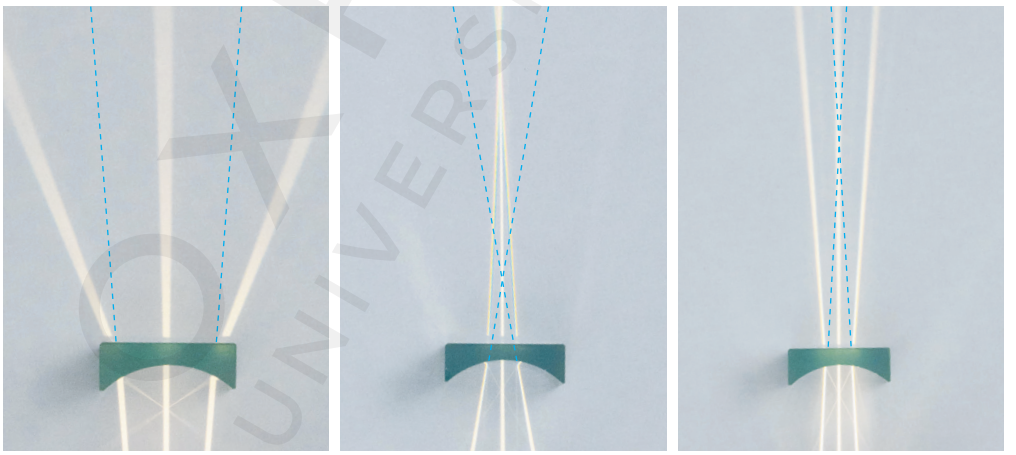


Convergent rays become more convergent.

Divergent rays become less divergent.

Divergent rays become convergent.

Fig 3.1j *Examples of convex lenses bending light rays towards the principal axis.*



Divergent rays become more divergent.

Convergent rays become less convergent.

Convergent rays become divergent.

Fig 3.1k *Examples of concave lenses bending light rays away from the principal axis.*

Right and wrong

✗ Light rays emerging from a convex lens always converge to a point.

✓ Light rays emerging from a convex lens **are more convergent than** the incident rays. However, if the incident rays are very divergent, the emerging rays may still be divergent.

Similarly, light rays emerging from a concave lens may still be convergent if the incident rays are very convergent.

Quick check 1

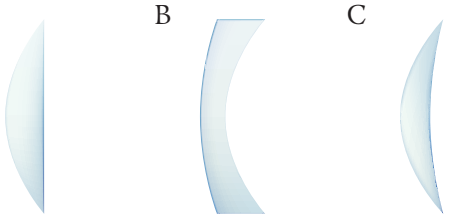
E-version in OUP Exercise Platform is also available.

Google Forms

Microsoft Forms

1 Which of the following lenses is **not** a convex lens?

A B C

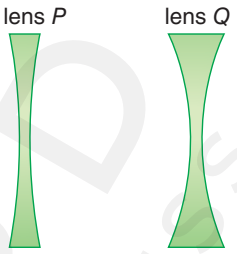


2 *True or false:*

(a) If a ray of light passes through the optical centre of a lens, it does not change direction. (T/F)

(b) The focal lengths on the two sides of a lens always have the same value. (T/F)

3 Lenses *P* and *Q* are made of the same material. Which lens has a shorter focal length?



4 Lenses *P* and *Q* have the same shape, but *P* is made of a material with a larger refractive index. Which lens has a shorter focal length?

Exercise 3.1

1 Which of the following statements about a convex lens is/are correct?

(1) It is a diverging lens.

(2) It is thicker at the centre.

(3) It bends light rays by refraction.

A (1) only

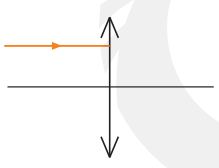
B (3) only

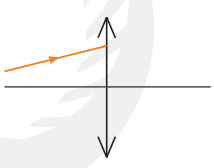
C (1) and (2) only

D (2) and (3) only

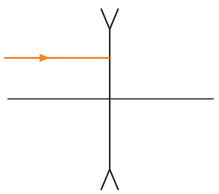
2 Sketch the emerging ray in each of the following cases.

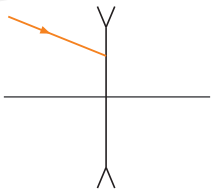
(a) (b)





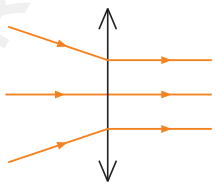
(c) (d)



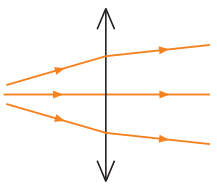


*3 Which of the following ray diagrams is/are possible? Put a tick in the corresponding box(es).

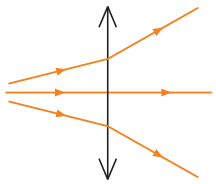
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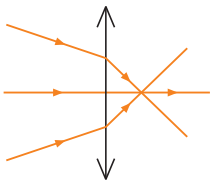
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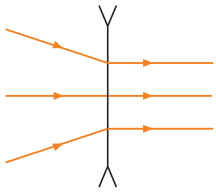


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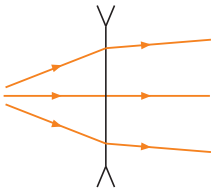


*4 Which of the following ray diagrams is/are possible? Put a tick in the corresponding box(es).

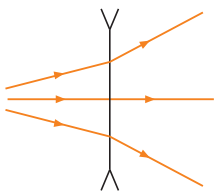
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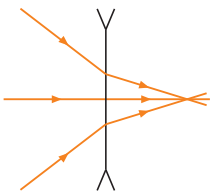
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3.2 Images formed by a convex lens

When you look at an object through a convex lens, you see the image formed by the lens. If the lens is placed at different distances from the object, the image will look different (Fig 3.2a).

In this part, we will learn about how we see the images formed by a convex lens and the nature of the images.

Before discussing the image properties, Ts may ask Ss to observe objects at different distances away through a convex lens. This allows Ss to experience the great diversity of images that can be formed by the lens



Fig 3.2a *Images formed by the same convex lens at different distances from the object.*

DSE goal

Distinguish between real and virtual images.

Flipped classroom

Watch a video and answer the questions.

1 Seeing the image

a Virtual images

Consider a small object *X* placed **within the focal length** of a convex lens. Light rays from *X* are very divergent when they hit the lens. As a result, the rays are still divergent after emerging from the lens (Fig 3.2b). All the emerging rays appear to come from the same point *Y*, which is on the same side of the lens as *X*.

When we look at *X* through the lens, we perceive the image at *Y*. The image is **virtual** because **no rays pass through it**.

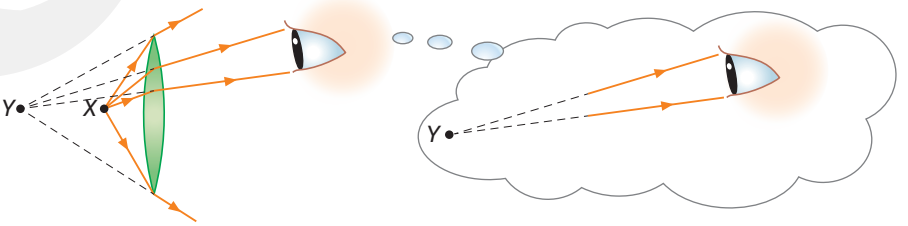


Fig 3.2b *Seeing a virtual image formed by a convex lens.*

We can see the image only when we observe it inside the yellow region (e.g. at *P* in Fig 3.2c). Outside this region (e.g. at *Q*), we cannot see the image because no emerging rays can enter our eyes.

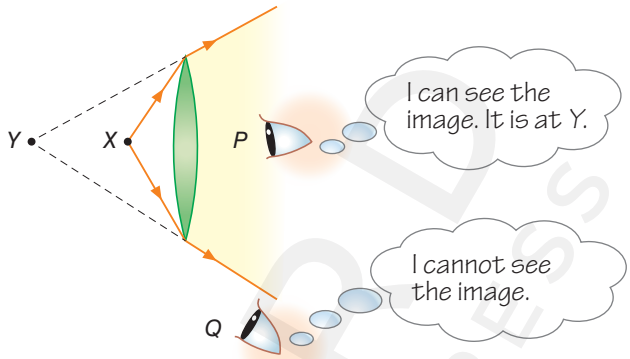


Fig 3.2c *We can see the image only when we observe it inside the yellow region.*

b Real images

Figure 3.2d shows a small object *X* placed **beyond the focal length** of a convex lens. Light rays from *X* converge to point *Y* after passing through the lens. When we look at *X* through the lens, we perceive the image at *Y*, which is on the other side of the lens. Such an image is called a **real image** since **light rays really diverge from it**.

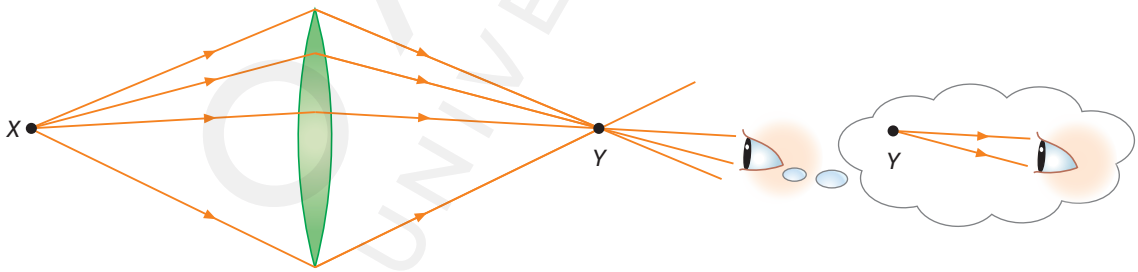


Fig 3.2d *Seeing a real image formed by a convex lens.*

We cannot see any image at positions nearer than *Y* to the lens. The rays entering our eyes are still converging, so they do not meet at a point when extended backwards.

We can see the image only when we observe it inside the yellow region (e.g. at *P* in Fig 3.2e). Outside this region (e.g. at *Q*), we cannot see the image since no emerging rays can enter our eyes.

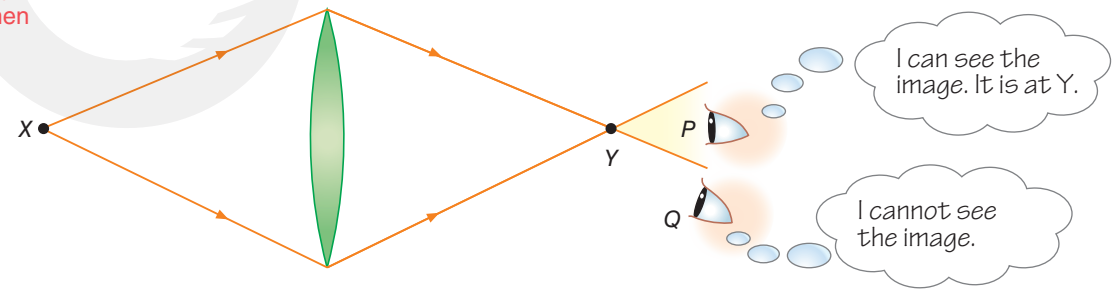


Fig 3.2e *We can see the image only when we observe it inside the yellow region.*

real image 實像

A real image can be projected on a screen. This allows us to see the image from outside the yellow region. To capture a sharp image, the screen should be put at the position where the emerging rays converge to.

In Figure 3.2f, a *translucent screen* is put at *Y*, so we can see the image from different directions.

After hitting the translucent screen, some rays undergo diffuse reflection and some pass through the screen.

Emphasize that light rays undergo diffuse reflection on the screen. Some rays from the image enter our eyes no matter where our eyes are.

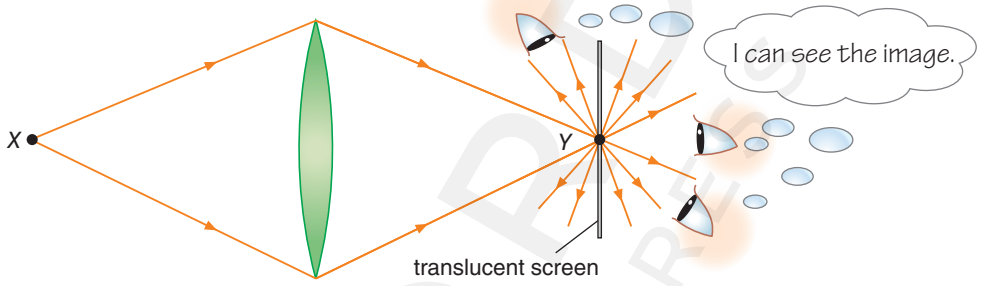


Fig 3.2f *Seeing a real image captured on a translucent screen.*

Cross reference

Our eyes also use convex lenses to form images. See Book E4 Chapter 1.

Projectors work by a similar principle. They use convex lenses to form real images on different surfaces (Fig 3.2g).

Unlike real images, **virtual images cannot be captured on a screen**.

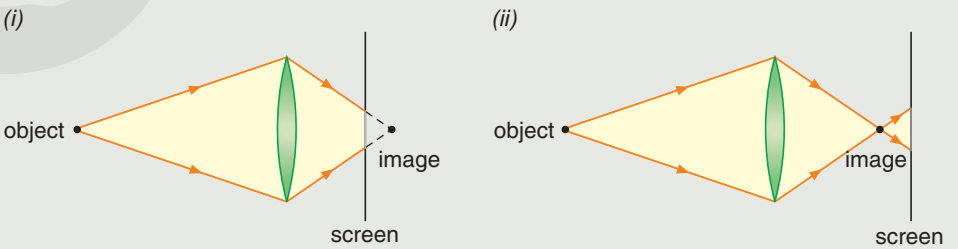


Fig 3.2g *Real images formed by projectors.*

Know more

Position of screen

If the screen is moved slightly away from the position of the image, the captured image becomes blurred. This is because a point on the object is projected as a small patch of light on the screen.



translucent screen 半透明屏幕

→ This simulation shows how images are formed by a convex lens.

Simulation Video

→ This video demonstrates Expt 3b. The image formed by the convex lens is captured on a translucent screen. This shows that the image is in mid-air and in front of the lens rather than behind or on the surface of the lens as some Ss may think.

Do not touch the hot ray-box lamp.

→ This is a Video Quiz about another experiment investigating the images formed by a convex lens.

Let us verify the above discussion in the following experiment.

Experiment 3b / Observing images formed by a convex lens

Use a convex lens of focal length 15–30 cm.

1 Put an illuminated letter 'F' in front of a convex lens (Fig a). The distance between them should be about two times the focal length of the lens.

2 Look at the lens from different positions. Can you observe the image?

3 Move a translucent screen to capture a sharp image of letter 'F' (Fig b). Can you observe the image from different positions?

4 Repeat steps 2 and 3 by placing letter 'F' within the focal length of the lens.



Fig a

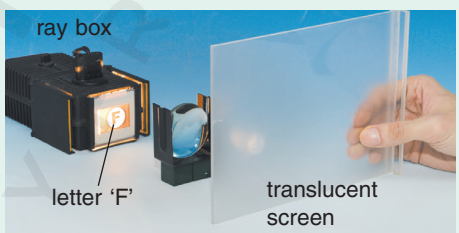


Fig b

Discussion

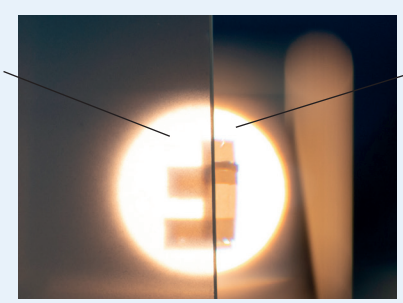
Can you capture a sharp image on the screen in step 4?

A convex lens can form virtual images and real images.
No light rays pass through a virtual image. The image cannot be projected on a screen.
Light rays actually pass through a real image. The image can be projected on a screen.

Right and wrong

✗ A screen or a flat surface is necessary for a real image to be formed.

✓ A real image does not need a screen to be formed. You can see the real image by looking at the lens directly.

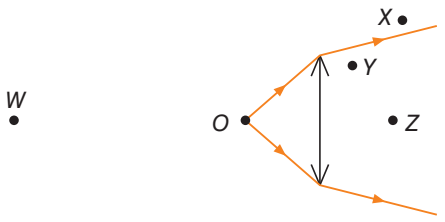


Google Forms Microsoft Forms

Quick check 2

E-version in OUP Exercise Platform is also available.

1 An object O is in front of a convex lens.



(a) Is the image of O formed by the lens real or virtual?

(b) At which position(s) shown can an observer see the image of O?

2 *True or false:*

(a) A convex lens can only form real images. (T/F)

(b) A virtual image cannot be captured on a screen. (T/F)

(c) There is light passing through a real image. (T/F)

DSE goal

Construct images formed by convex lens.

DSE exam

14(1B)Q6, 18(1B)Q6a, 19(1A)Q20, 20(1A)Q20, 22(1A)Q21, 22(1B)Q5

2 Construction and nature of images

a Construction rules

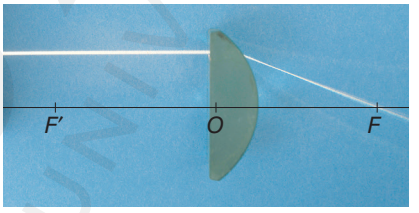
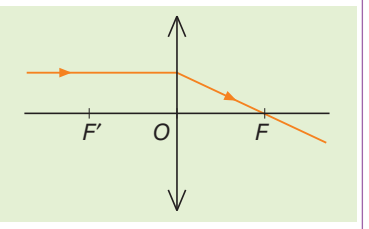
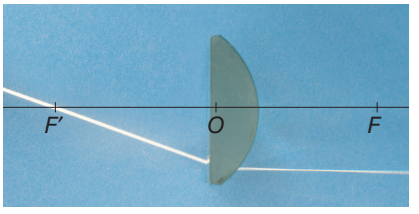
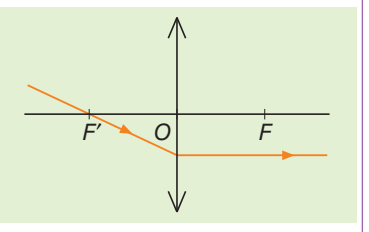
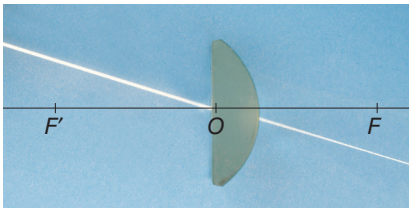
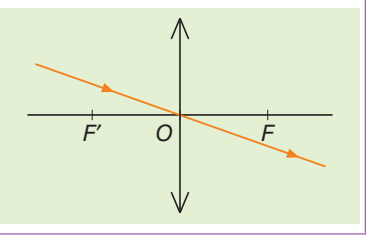
Stress that the construction rules refer to rays along some specific directions. Rays along other directions can also be used to draw a ray diagram but this would be more complicated.

We can draw a ray diagram to locate the image formed by a convex lens. Although the bending of light rays by a lens follows the laws of refraction, it is in practice impossible to trace the rays by this means. Instead, we use rays along certain specific directions as the construction rules.

Video

→ This video shows how a ray of light from a ray box is refracted by a convex lens.

Construction rules for a convex lens

Rule 1 A ray parallel to the principal axis is bent so that it passes through the principal focus F on the other side of the lens.		
Rule 2 A ray passing through the principal focus F' emerges parallel to the principal axis.		
Rule 3 A ray passing through the optical centre travels straight on.		

Cross reference

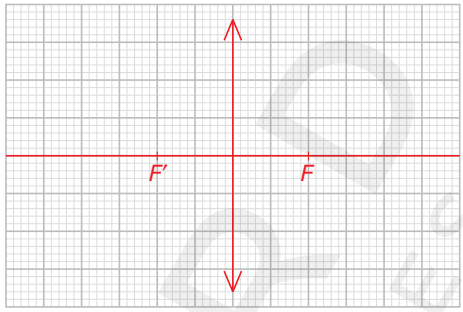
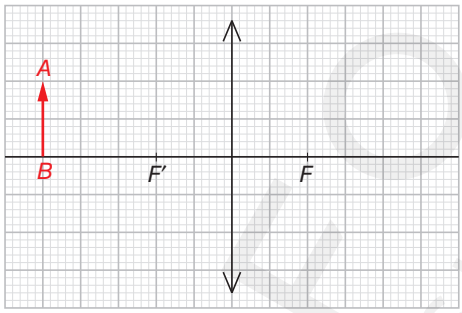
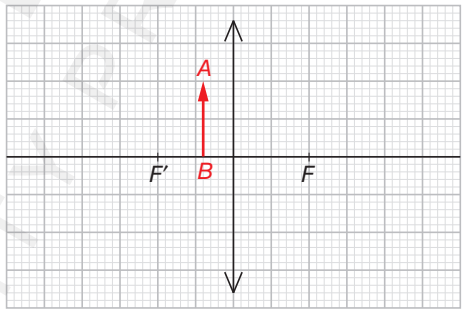
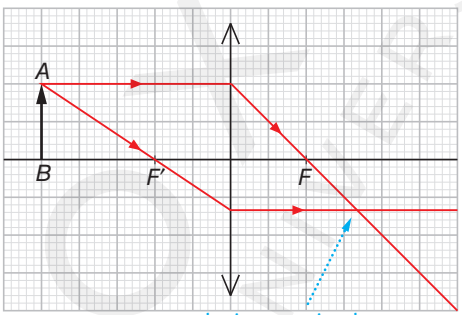
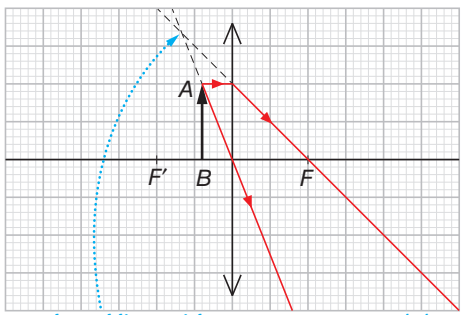
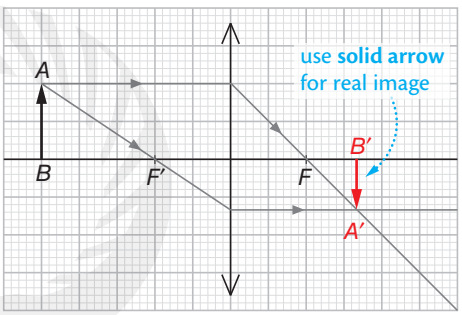
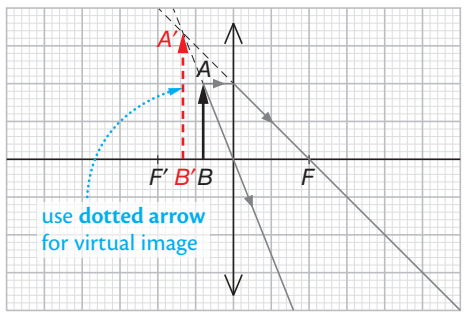
See p.9 to review the meaning of reversibility of light.

▶ Recalling the reversibility of light, we can see that rule 2 is just the reverse of rule 1.

→ This simulation shows how the construction rules are used to draw ray diagrams for a convex lens. Ss can move the object away from the principal axis and change the orientation of the object.

Simulation

With the three construction rules, a ray diagram can be drawn to determine the position and nature of the image formed by a convex lens.

Steps	Case 1 (object beyond F)	Case 2 (object within F)
Principal axis, lens and foci Draw the principal axis and the symbol of lens. Then mark the foci F and F'.		
Object Draw the object AB as an upright arrow with its tail B on the principal axis at the given position.		
Construction rules Draw rays from tip A using any two construction rules. Remind Ss that (a) the rays only bend at the vertical line of the symbol for the lens in a ray diagram; (b) only two out of the three construction rules are needed for locating an image.		
Image Draw the image as a vertical arrow with its tail on the principal axis. For a vertical object with its tail on the principal axis, its image must also be vertical with its tail on the principal axis.		

▲ A ray travels straight on along the principal axis. If the tail of the object lies on the principal axis, the tail of the image will also lie on the principal axis.

After the illustration, Ts may ask Ss 'What happens if the object AB is at other positions?'

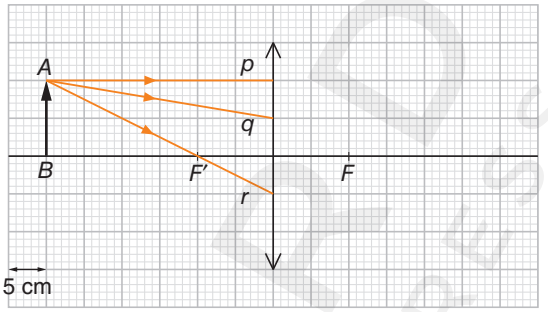
The distance of the object from the lens is called the **object distance** (usually denoted by u) and the distance of the image from the lens is called the **image distance** (usually denoted by v).

Note that an object can only form one image, i.e. a certain object distance corresponds to a unique image distance.

This example reinforces the idea that rays from the object can be directed at the lens along directions other than those of the construction rules. All rays from an object that are refracted by the convex lens will converge to, or appear to come from, the image *I*.

Example 1 | Image formed by a convex lens (I)

An object *AB* is in front of a convex lens with foci *F* and *F′*. Light rays *p*, *q* and *r* coming from point *A* falls on the lens as shown.

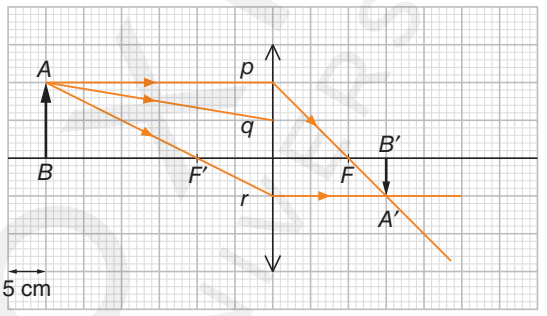


(a) Locate the image *A′B′* by completing rays *p* and *r*. What is the image distance?

(b) Hence, complete ray *q*.

Solution

(a)



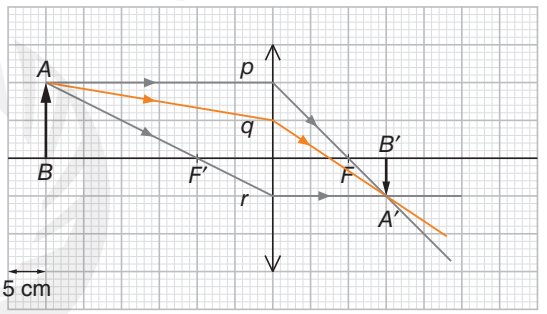
The image distance is 15 cm.

Tip

Ray *q* is not included in the construction rules. However, since it comes from *A*, it must pass through *A′*.

Similarly, a ray from *B* must pass through *B′*.

(b)

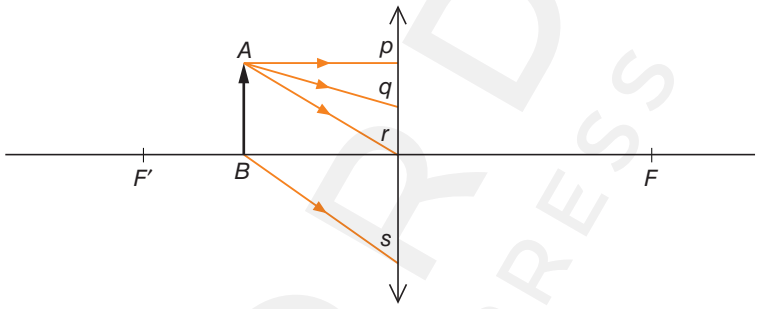


Think about...

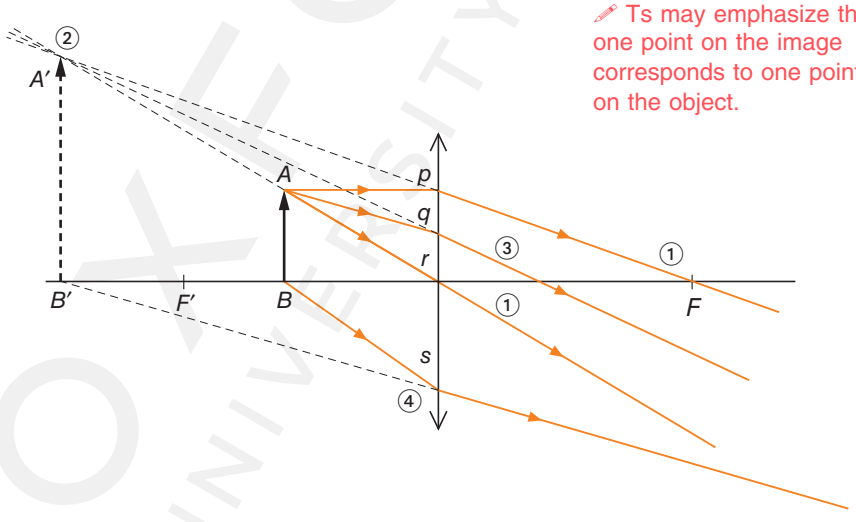
The convex lens is replaced by one with a focal length of 15 cm. Suppose positions of the object and the lens are unchanged. Locate the new image of the object.

Example 2 Image formed by a convex lens (II)

An object AB is in front of a convex lens with foci F and F'. Light rays p, q, r and s coming from AB falls on the lens as shown. Sketch the image and the emerging rays.



Solution



Ts may emphasize that one point on the image corresponds to one point on the object.

Tip

① Complete rays p and r according to the construction rules.

② Locate the image $A'B'$ by extending the rays in ① backwards.

③ Since ray q comes from A, it appears to come from A' after passing through the lens.

④ Since ray s comes from B, it appears to come from B' after passing through the lens.

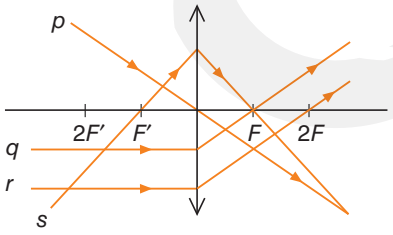
Quick check 3

E-version in OUP Exercise Platform is also available.

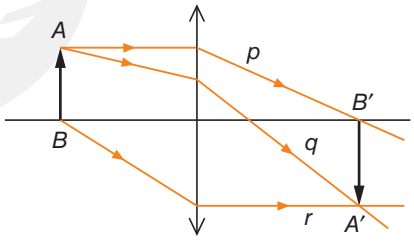
Google Forms Microsoft Forms

1 Which of the following light rays is/are drawn correctly? In (b) and (c), $A'B'$ is the image of AB.

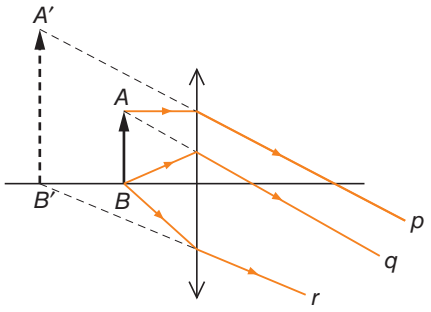
(a)



(b)



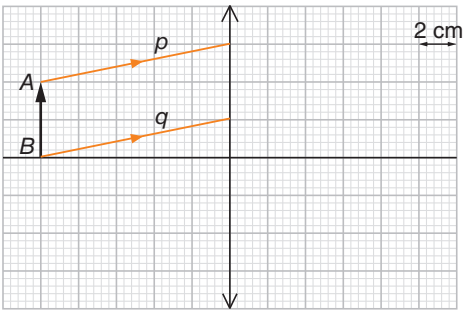
(c)



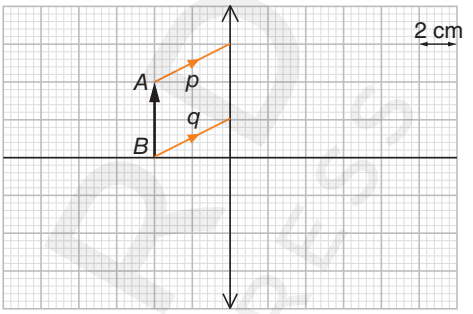
Learn by practice 1

1 In each of the following cases, use the construction rules to locate the image of AB. Hence, complete rays p and q.

(a) The focal length of the lens is 4 cm.



(b) The focal length of the lens is 10 cm.



Recall that parallel rays always converge to a point after passing through a convex lens. If the incident rays are parallel to the principal axis, they will converge to the focus F (Fig 3.2h). Otherwise, the parallel rays will converge to another point on the focal plane.

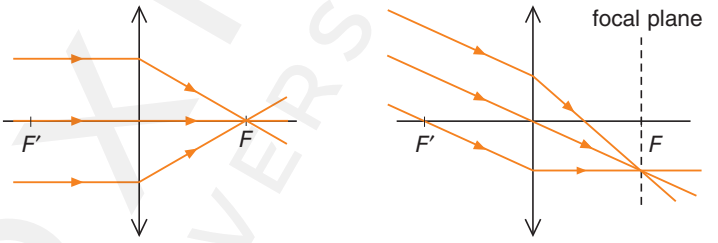


Fig 3.2h *A convex lens converges parallel rays on the focal plane.*

Light rays from a point on a distant object are parallel.

For a lens of focal length about 10 cm, an object 10 m away can be regarded as a distant object.

▶ If a screen is placed on the focal plane, it can capture the image of distant objects. By using this property, we can measure the focal length f of a convex lens (Fig 3.2i).

1 Use the lens to project the image of a distant object on a screen.

2 The distance between the lens and the screen is f.

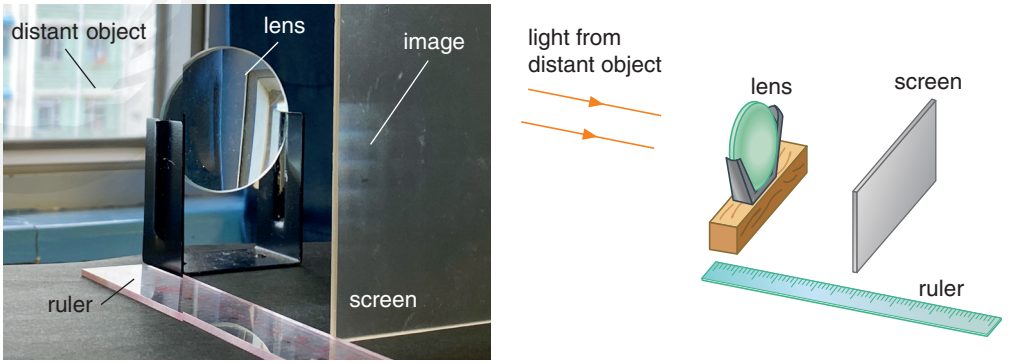


Fig 3.2i *Measuring the focal length of a convex lens.*

Know more

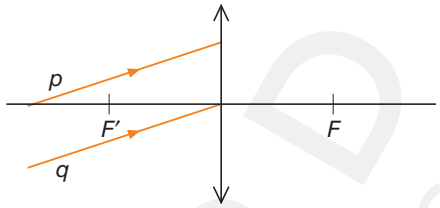
Measuring *f*

We can also measure the focal length of a convex lens by the help of a mirror.

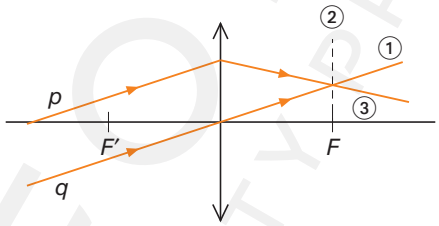
▶ p.132

Example 3 Parallel rays through a convex lens

Parallel rays p and q hit a convex lens as shown. Sketch the emerging rays.



Solution



Tip

① According to rule 3, q travels straight on.

② Draw the focal plane.

③ Since p and q are parallel before passing through the lens, they must intersect on the focal plane.

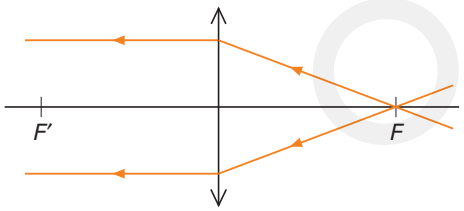
Quick check 4

E-version in OUP Exercise Platform is also available.

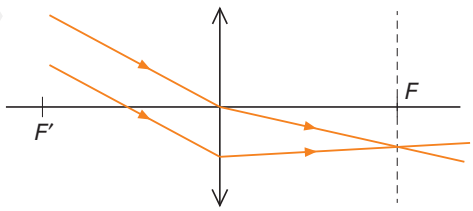
Google Forms Microsoft Forms

1 Determine whether each of the following ray diagrams is correctly drawn.

(a)

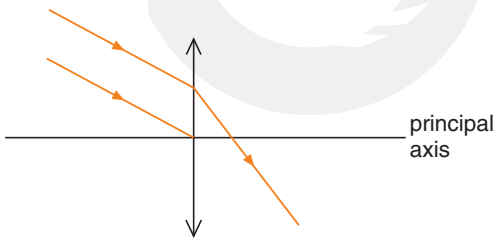


(b)

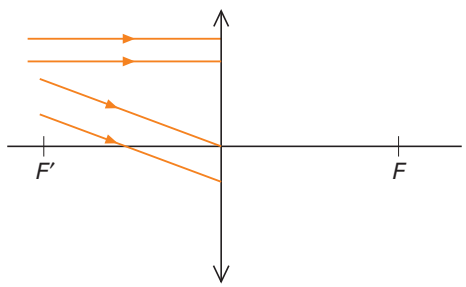


Learn by practice 2

1 Locate the focus F of the lens below.



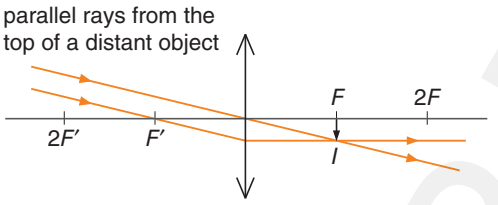
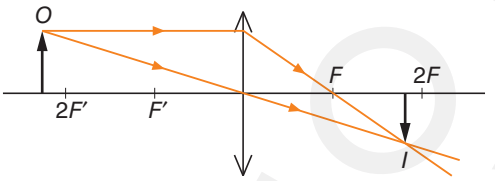
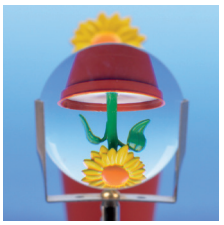
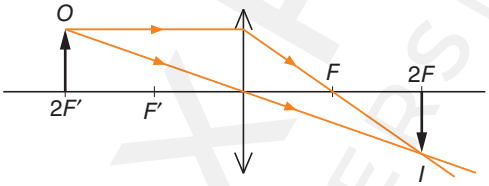
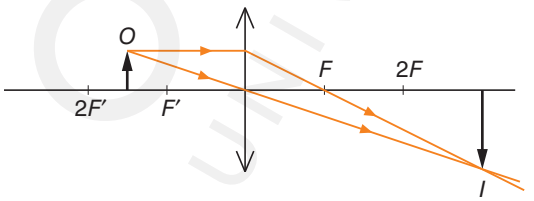
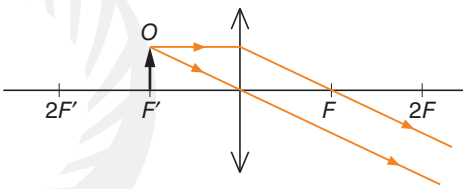
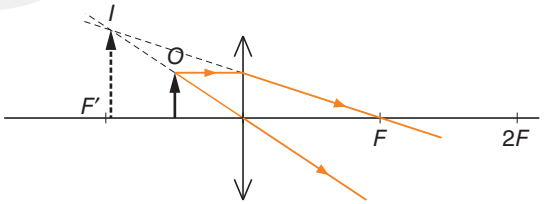
2 Complete the rays below.



✎ Ts may need to emphasize that 'inverted' or 'erect' does not refer to the absolute orientation of image. It is a comparison between the orientation of the object and the image.

b Position and nature of images

The nature of the image *I* changes with the position of the object *O*.

Position of *O*	Ray diagram	Position of *I*	Nature of *I*
at infinity (distant object) 		on focal plane	real inverted diminished
beyond 2*F*′ 		between *F* and 2*F*	real inverted diminished
at 2*F*′ 		at 2*F*	real inverted same size
between 2*F*′ and *F*′ 		beyond 2*F*	real inverted magnified
at *F*′ 		no image formed (or image formed at infinity)	
within *F*′ 		same side as the object	virtual erect magnified

✎ Real images formed by lenses must be inverted while virtual images must be erect. Real images are always formed on the opposite side of the lens while virtual images are always formed on the same side as the object.

We can verify the above by doing Experiment 3c.

→ This is a Video Quiz about this experiment.

→ This video demonstrates Expt 3c.

Video

Do not touch the hot ray-box lamp.

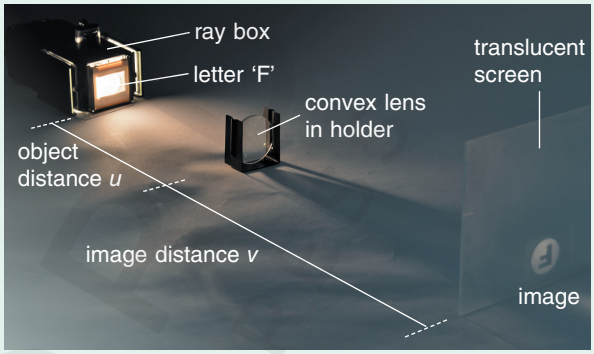
Experiment 3c Nature of images formed by a convex lens

1 Put a convex lens in front of an illuminated letter 'F'.

2 Observe the nature of the image of the letter.

3 Try to capture the image of the letter on a screen.

4 Repeat steps 2 and 3 with different object distances.



Discussion

How does the image change when the object distance u changes? See p.96.

When an object is at a distance u from a convex lens of focal length f,

- **if $u > f$, a real and inverted image is formed on the opposite side of the object;**
- **if $u < f$, a virtual, erect and magnified image is formed on the same side as the object.**

Ts may need to emphasize that 'inverted' or 'erect' does not refer to the absolute orientation of image. It is a comparison between the orientation of the object and the image.

Know more

Cylindrical and spherical convex lenses

Do you know how the images formed by these two kinds of convex lenses differ?

▸ p.132

Know more

Convex lenses in daily life

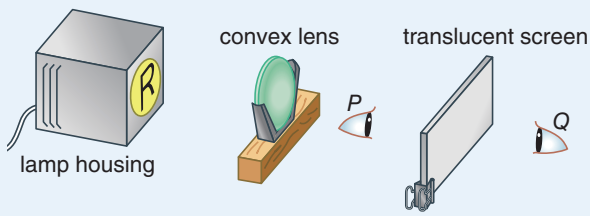
Convex lenses can easily be found in daily life. For example, they are commonly used as magnifying glasses.



The human eye has a convex lens to project real images onto the *retina*. Furthermore, convex lenses can be used to correct long sight. This will be further discussed in Book E4.

Right and wrong

A real image of the letter 'R' is formed on a translucent screen. Does the image observed by P and Q look the same?



✗ Both P and Q can see an inverted image, so the images observed by them are the same.

✓ No, P and Q observe the image from different sides of the screen. The image seen by P and Q look like this:

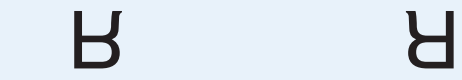


image seen by P

image seen by Q (look the same as if we look at the lens without using a screen)

retina 視網膜

Quick check 5

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) The virtual images formed by a convex lens must be magnified. (T/F)

(b) If an image formed by a convex lens is magnified, it must be virtual. (T/F)

2 A glass ball forms an image like a convex lens does (Fig a).



Fig a

(a) Can this image be projected on a screen?

(b) Is this image formed on the focal plane of the glass ball?

3 A man puts a convex lens of focal length f in front of his eye. An image of his eye is formed (Fig b).



Fig b

(a) Is the image real?

(b) Is the distance between the lens and the man shorter or longer than f?

(c) Will the image become inverted eventually if he moves the lens away from his eye?

c Linear magnification

Linear magnification is called magnification for short.

▶ The **linear magnification** m of an image is defined as (Fig 3.2j):

height of image (cm) → h_i

linear magnification (no unit) → $m = \frac{h_i}{h_o} = \frac{v}{u}$

height of object (cm) → h_o

v → **image distance** (cm)

u → **object distance** (cm)

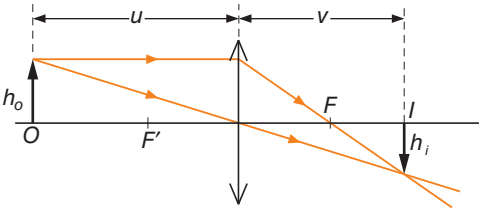


Fig 3.2j Linear magnification of the image formed by a convex lens.

Linear magnification has no unit.

When $m > 1$, the image is magnified.

When $m = 1$, the image is of the same size as the object.

When $m < 1$, the image is diminished.

linear magnification 線性放大率

Example 4 Magnification of an image formed by a convex lens

An object of height 5 cm is placed at 15 cm from a convex lens of focal length 10 cm.

(a) Find the position and magnification of the image.

(b) How do the magnification change when the object is moved away from F'?

Solution

(a)

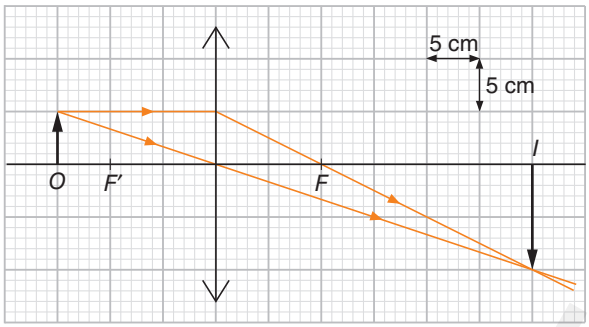
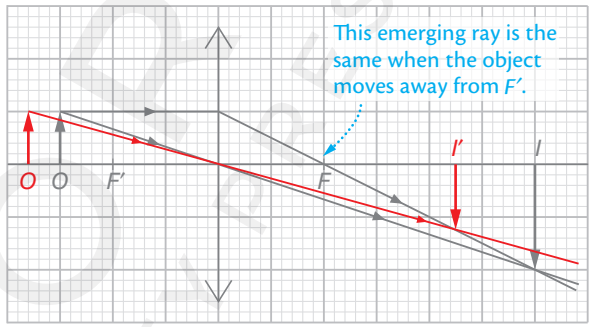


Image distance $v = 30$ cm

$m = \frac{v}{u} = \frac{30}{15} = 2$

(b)



The magnification decreases.

Note that the magnification **decreases** when the object is **further away from the focus**. This is true for both real and virtual images.

Quick check 6

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) A virtual image formed by a convex lens must have a linear magnification larger than 1. (T/F)

(b) When the linear magnification of an image formed by a convex lens is smaller than 1, the image distance must be smaller than the object distance. (T/F)

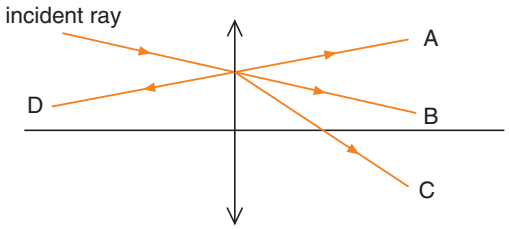
(c) When an object is moved from a point very close to a convex lens towards the focus of the lens, the magnification of the image increases. (T/F)

2 Complete the table below.

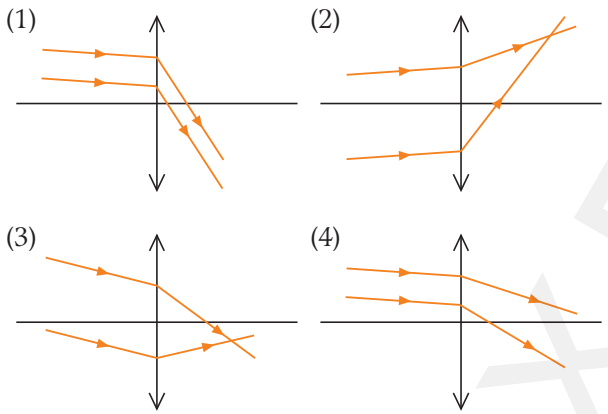
	u / cm	v / cm	h_o / cm	h_i / cm	m
(a)	20	5		2	
(b)		30	10		3

Exercise 3.2

1 Which light ray in the figure below could represent the refracted ray?



***2** Which of the following ray diagrams is possible for a pair of parallel incident rays refracted by a convex lens?



A (3) only
B (2) and (3) only
C (1) and (4) only
D (1), (2), (3) and (4)

***3** A sharp image is captured on the translucent screen as shown in Figure a. Which of the following is correct if the screen is removed?

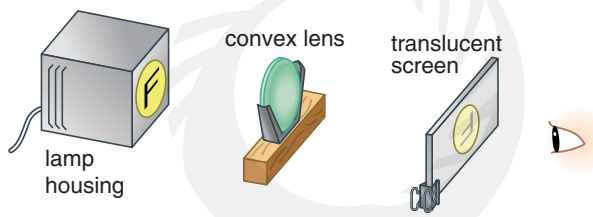


Fig a

A The image becomes virtual.
B The image remains at the same position.
C The observer can no longer see the image.
D The observer must move to the position where the screen was to see the image.

***4** An object is placed in front of a convex lens (Fig b). Which observer(s) can see the object's image formed by the lens?

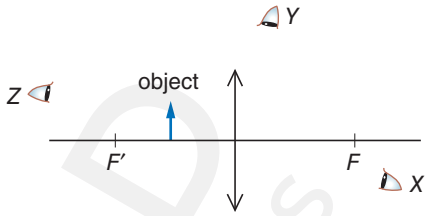


Fig b

A *X* B *Z*
C *X* and *Y* D *X*, *Y* and *Z*

***5** Figure c shows four rays *w*, *x*, *y* and *z* from an object *O* incident on convex lens *L*.

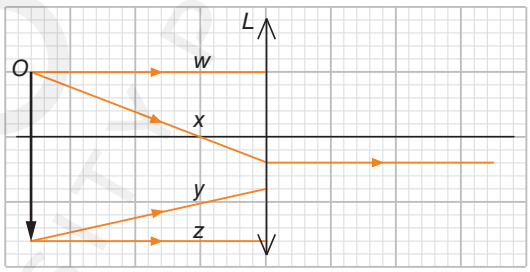


Fig c

(a) Mark and label the two foci of *L* in Figure c.
(b) Complete the paths of the rays, and locate the image *I*. [Hint: Use *w* to locate the tail of the object.]
(c) Is it a virtual or a real image? Real

***6** Figure d shows the image of a wooden dummy formed by a glass ball. The magnification of the image is 0.5.

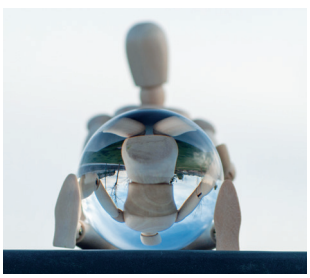


Fig d

(a) Is the image formed by the glass ball virtual or real? Explain your answer. Real
(b) The dummy is 10 cm tall and is 18 cm from the glass ball.
(i) Find the image distance. 9 cm
(ii) Find the height of the image. 5 cm

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3.3 Images formed by a concave lens

When you look at an object through a concave lens, you can always see an erect and diminished image (Fig 3.3a). Is the image real or virtual?



Fig 3.3a *Images formed by a concave lens.*

Flipped classroom
Watch a video and answer the questions.

1 Seeing the image

Consider a small object *X* placed in front of a concave lens. Since the lens is diverging, rays from *X* become more divergent and appear to come from point *Y* after passing through the lens. When we look at *X* through the lens, we perceive the image at *Y* (Fig 3.3b). The image is **virtual** since no light actually pass through it.

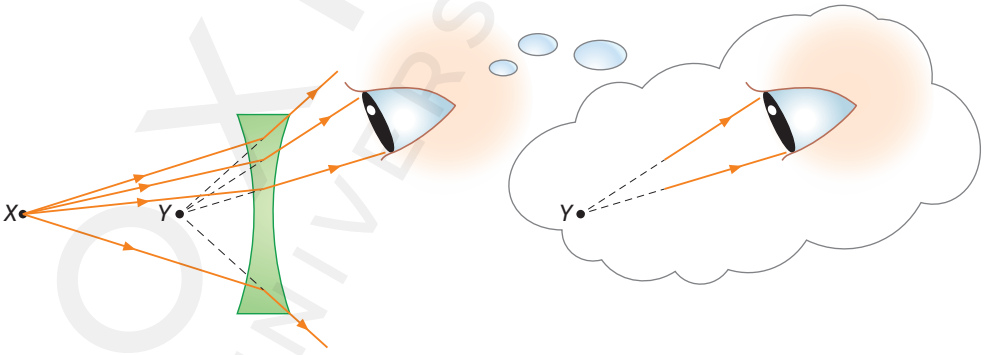


Fig 3.3b *Seeing a virtual image formed by a concave lens.*

We can see the image only when we observe it inside the yellow region (e.g. at *P* in Fig 3.3c). Outside this region (e.g. at *Q*), we cannot see the image because no emerging rays can enter our eyes.

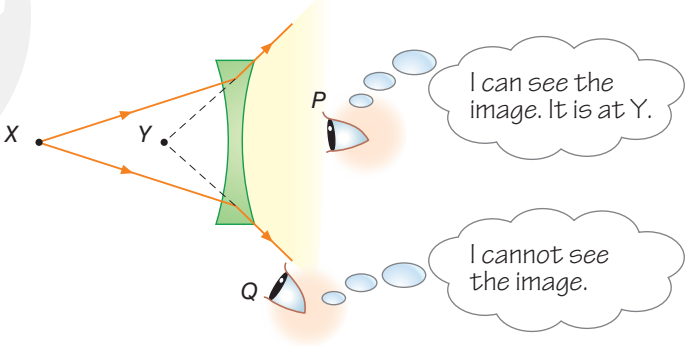


Fig 3.3c *We can see the image only when we observe it inside the yellow region.*

→ This simulation shows how a cone of rays from a point on the object is refracted by a concave lens.

Simulation
Video

→ This video demonstrates Expt 3d.

Do not touch the hot ray-box lamp.

Experiment 3d / Observing images formed by a concave lens

1 Put an illuminated letter 'F' in front of a concave lens (Fig a).
2 Look at the lens from different positions.
3 Try to capture the image on a translucent screen (Fig b).
4 Repeat steps 2 and 3 with different object distances.



Fig a



Fig b

Discussion

Can you capture a sharp image on the screen? No

A concave lens forms only virtual images.

DSE goal

Construct images formed by concave lens.

DSE exam

12(1A)Q21, 15(1A)Q15, 19(1B)Q6, 20(1A)Q18, 21(1A)Q18, 23(1A)Q20

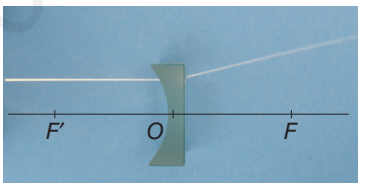
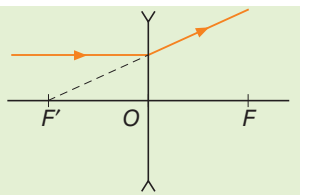
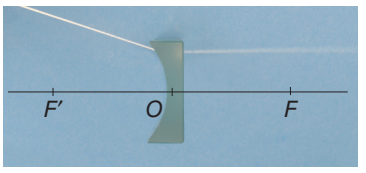
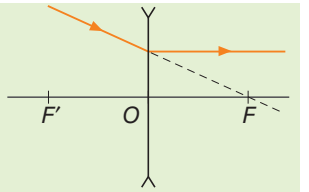
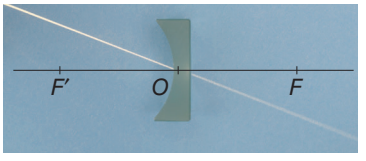
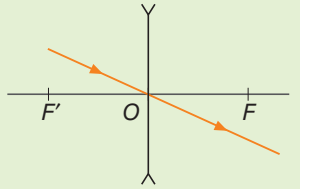
Video

→ This video shows how a ray of light from a ray box is refracted by a concave lens.

2 Construction and nature of images

a Construction rules

The following are the construction rules for a concave lens.

Construction rules for a concave lens		
Rule 1 A ray parallel to the principal axis is bent so that it appears to come from the principal focus *F′*.		
Rule 2 A ray directed towards the principal focus *F* emerges parallel to the principal axis.		
Rule 3 A ray passing through the optical centre travels straight on.		

Similar to the case of convex lenses, the construction rules refer to rays along some specific directions. Rays along other directions can also be used to draw a ray diagram but this would be more complicated.

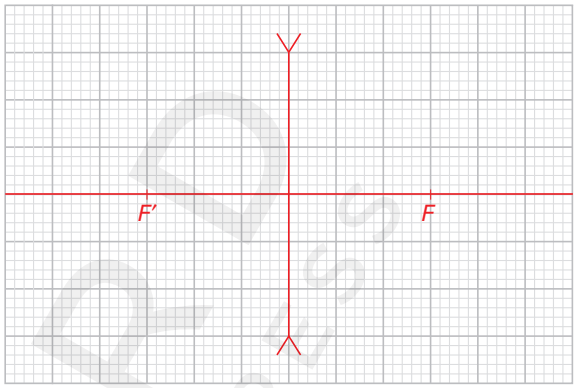
Simulation

→ This simulation shows how the construction rules are used to draw ray diagrams for a concave lens.

We can now draw a ray diagram to find the position and nature of the image formed by a concave lens.

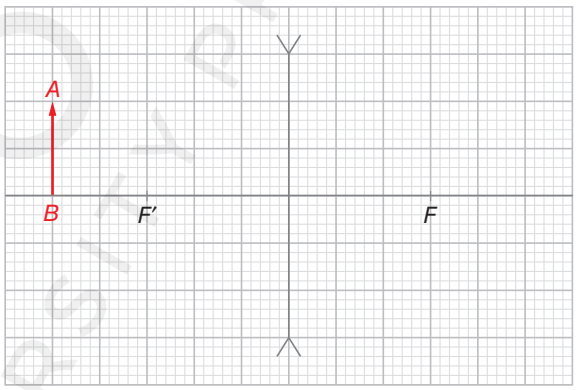
Principal axis, lens and foci

Draw the principal axis, the symbol of lens and the foci F and F'.



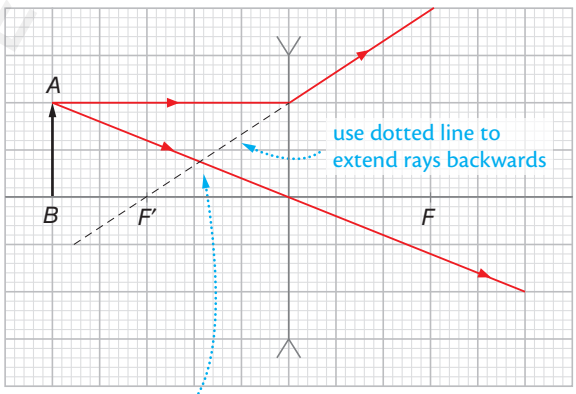
Object

Draw the object AB as an upright arrow with its tail B on the principal axis at the given position.



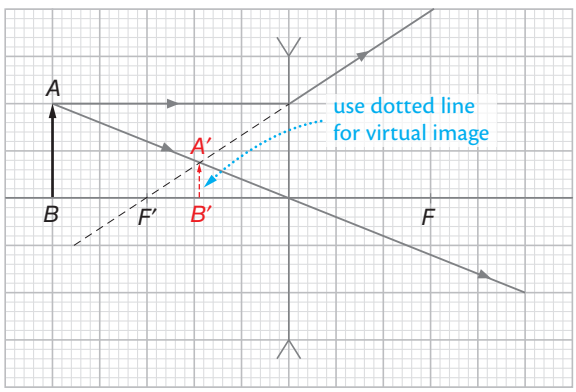
Construction rules

Draw rays from tip A using any two construction rules.



Image

Draw the image as a vertical arrow with its tail on the principal axis.

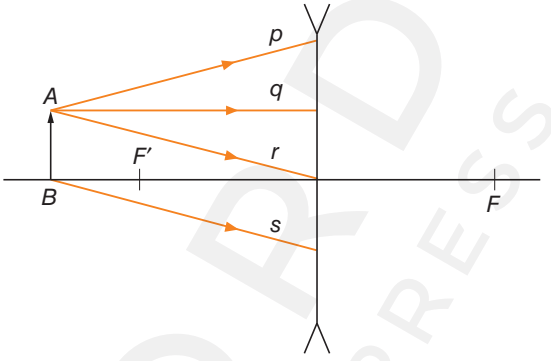


Remind Ss that (a) the rays only bend at the vertical line of the symbol for the lens in a ray diagram; (b) only two out of the three construction rules are needed for locating an image.

After the illustration, Ts may ask Ss 'What happens if the object AB is at other positions?'

Example 5 Image formed by a concave lens

An object AB is placed in front of a concave lens. Light rays p, q, r and s coming from the object falls on the lens as shown.

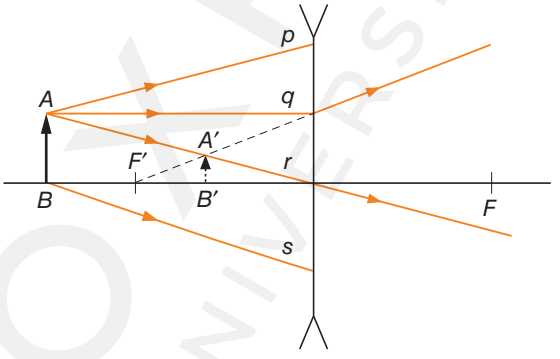


(a) Locate the image $A'B'$ by completing rays q and r.

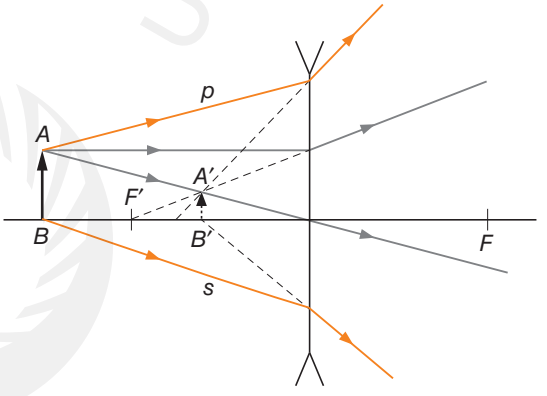
(b) Hence, complete rays p and s.

Solution

(a)



(b)



Since ray p comes out from A, it appears to come from A' after passing through the lens. Similarly, ray s comes from B and appears to come from B' after passing through the lens.

This example reinforces the idea that rays from the object can be directed at the lens along directions other than those of the construction rules. All rays from an object that are refracted by the concave lens will appear to diverge from the virtual image I.

Similar to the case of convex lenses, one point on the image corresponds to one point on the object.

Think about...

How will the image distance change if the lens is replaced by another one with a longer focal length?

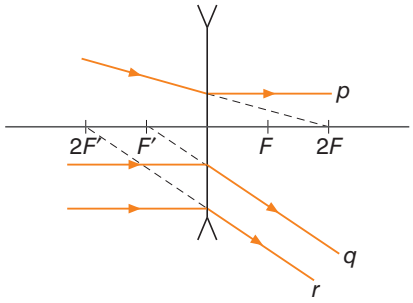
Quick check 7

E-version in OUP Exercise Platform is also available.

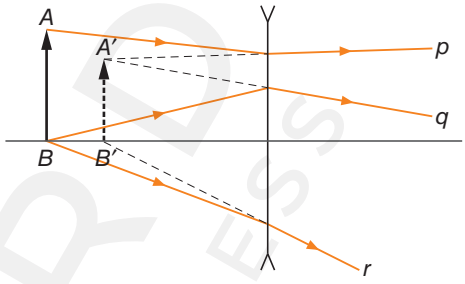
Google Forms

Microsoft Forms

1 Which of the following light rays is/are drawn correctly?

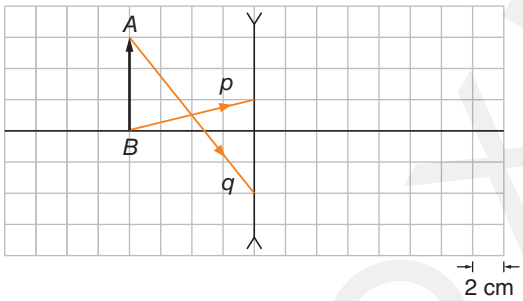


2 Object AB is put in front of a concave lens and image $A'B'$ is formed. Which of the following rays is/are drawn correctly?



Learn by practice 3

1 Object AB is 8 cm away from a concave lens. The focal length of the lens is 10 cm.

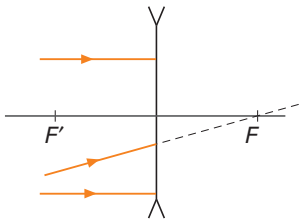


(a) Mark F and F' on the figure.

(b) Using any two construction rules, locate the image of AB.

(c) Hence, complete rays p and q.

2 Complete the following rays.



When parallel rays pass through a concave lens, they appear to diverge from a point (Fig 3.3d). If the incident rays are parallel to the principal axis, they will appear to diverge from the focus F'. Otherwise, the parallel rays will appear to diverge from another point on the focal plane.

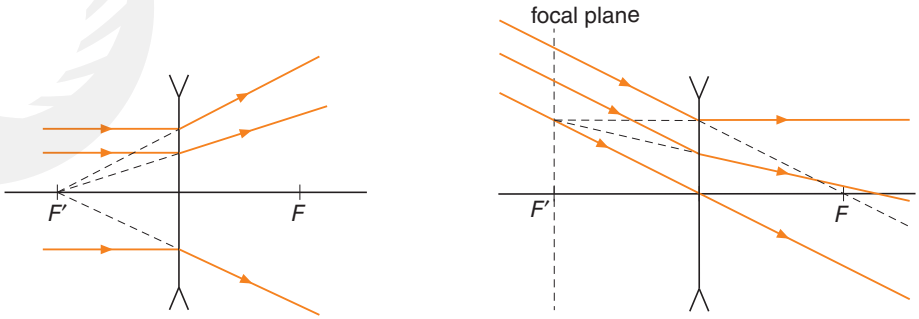
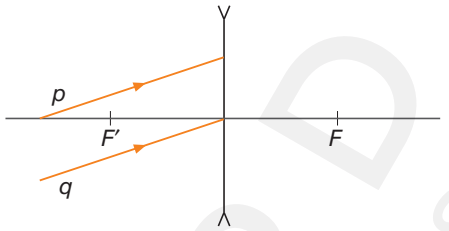


Fig 3.3d *Parallel rays appear to diverge from a point on the focal plane after passing through a concave lens.*

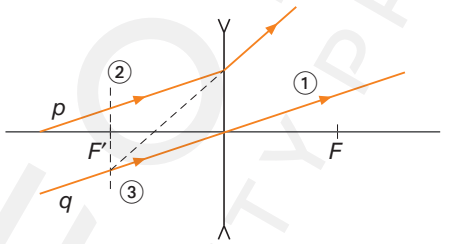
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Example 6 Parallel rays through a concave lens

Parallel rays p and q hit a concave lens as shown. Sketch the emerging rays.



Solution



Tip

① According to rule 3, q travels straight on.

② Draw the focal plane.

③ Since p and q are parallel before passing through the lens, they appear to come from the same point on the focal plane after passing through the lens.

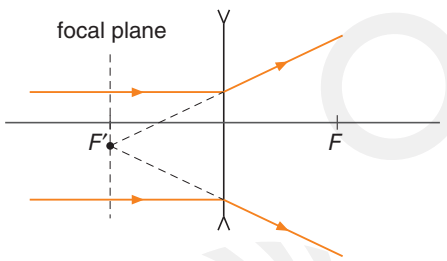
Quick check 8

E-version in OUP Exercise Platform is also available.

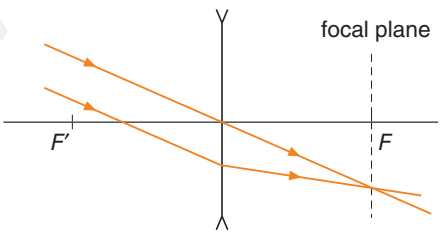
Google Forms Microsoft Forms

1 Determine whether each of the following ray diagrams is correctly drawn.

(a)

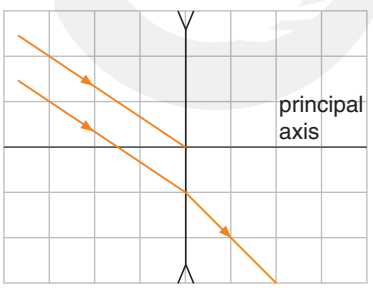


(b)

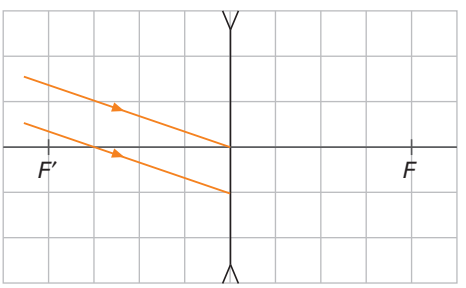


Learn by practice 4

1 Locate the focus on the left side of the lens.



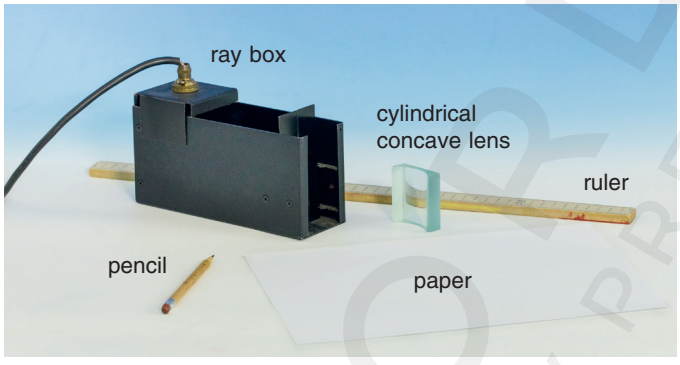
2 Complete the rays below.



Writing practice

Question

You are given a ray box that produces parallel light rays, a cylindrical concave lens, a large piece of paper, a pencil and a ruler.



Describe how you can find the focal length of the lens using the above apparatus. Suggest a source of error in the experiment. (5 marks)

Answer

Put the lens on the paper. Trace its outline. 1A

Direct the parallel light rays to the lens. Trace the path of the rays. 1A

Remove the lens and the ray box. Extend the emerging rays backwards. The intersection of extended rays lies on the focal plane. 1A

The focal length of a lens should be the perpendicular distance from the intersection to the lens. 1A

Source of error (any one): 1A

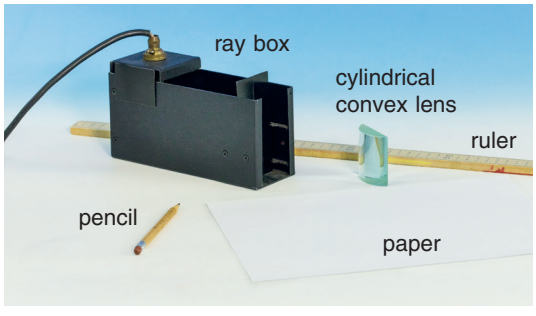
The path of the ray is not marked correctly due to its thickness.

The central part of the lens is not located correctly.

Practise yourself

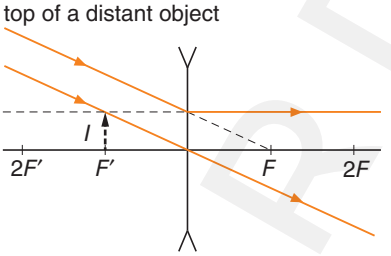
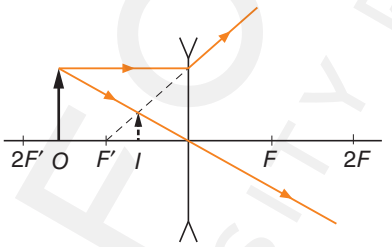
You are given a ray box that produces parallel light rays, a cylindrical convex lens, a large piece of paper, a pencil and a ruler.

Describe how you can find the focal length of the lens using the above apparatus. Suggest a source of error in the experiment. (4 marks)



b Position and nature of images

The table below summarizes how the nature of the image *I* formed by a concave lens changes when the object *O* is at different positions.

Position of *O*	Ray diagram	Position of *I*	Nature of *I*
at infinity (distant object) 	parallel rays from the top of a distant object 	on focal plane	virtual erect diminished
other positions 		within *F*	virtual erect diminished

Ts may emphasize that, for concave lens, image nature is the same for all object distance. That's why there is no need to consider different cases.

For a concave lens, a virtual, erect and diminished image is always formed on the same side as the object.

c Linear magnification

Same as with a convex lens, the linear magnification *m* of the image formed by a concave lens is given by $m = \frac{h_i}{h_o} = \frac{v}{u}$ (Fig 3.3e). Since the image is always diminished, *m* is always smaller than 1.

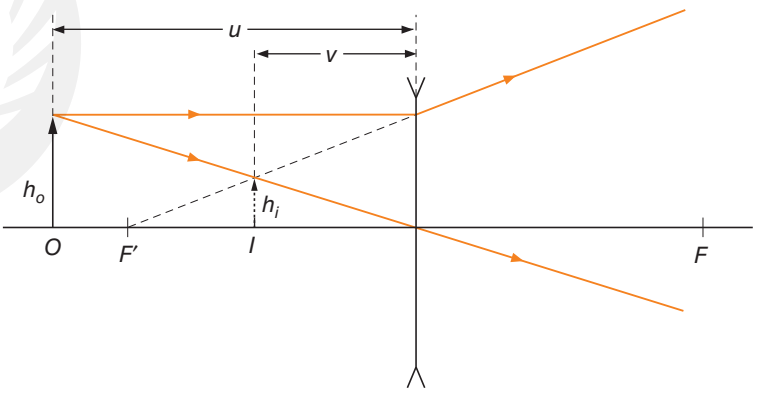


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Know more

Concave lenses in daily life

A concave lens can be used to correct short sight. It is also used as peephole lens as the images formed are always diminished (to provide a wider view) and erect.



Example 7 Magnification of an image formed by a concave lens

The figure shows the view through a concave lens of focal length 2 m. The lens is 2 m from the chair.



(a) State the nature of the image.

(b) Draw a ray diagram to locate the image of the chair.

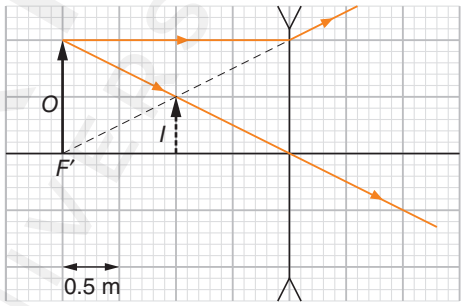
(c) Find the magnification of the image.

(d) If the lens is moved closer to the chair, how will the magnification of the image change?

Solution

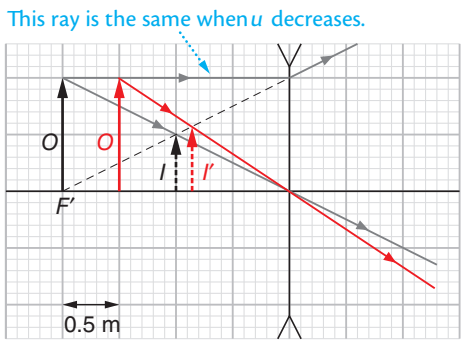
(a) The image is virtual, erect and diminished.

(b)



(c) $m = \frac{v}{u} = \frac{1}{2} = 0.5$

(d)



The image will be larger, so the magnification will increase.

Note that the image becomes **larger** when the object is **closer to the lens**, but the magnification is always smaller than 1.

Google Forms Microsoft Forms

Quick check 9

E-version in OUP Exercise Platform is also available.

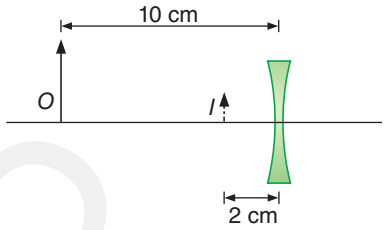
1 What is the magnification of the image?

A 0.2

B 0.5

C 2

D 5



Exercise 3.3

★**1** Which of the following statements about the image formed by a concave lens is/are correct?

(1) The image distance is always smaller than the focal length of the lens.

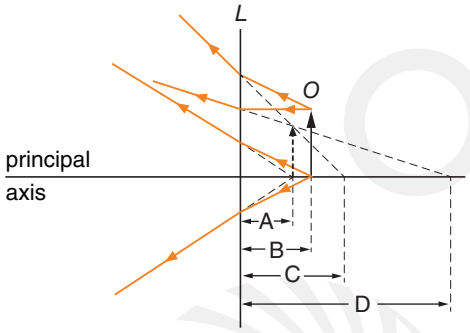
(2) It is inverted.

(3) It is always located on the same side of the lens as the object.

A (1) only B (3) only

C (1) and (3) only D (1), (2) and (3)

★**2** Which of the following indicates the focal length of lens *L*?



3 Draw a ray diagram to locate the image (Fig a). State the nature of the image.

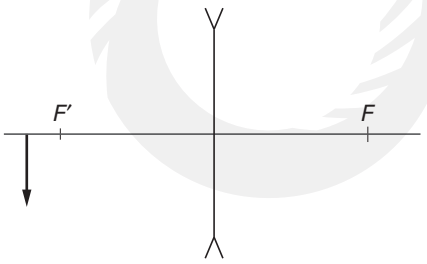


Fig a

★**4** An object *AB* is placed in front of a concave lens (Fig b).

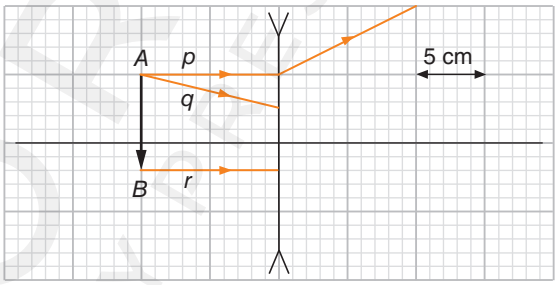


Fig b

(a) Locate the two foci in Figure b.

(b) Complete the paths of *q* and *r*, and locate the image of *AB*. [Hint: use *r* to locate the image of *B*.]

★**5** An object *O* is 12 cm from a concave lens (Fig c). The linear magnification of the image is $\frac{1}{3}$.

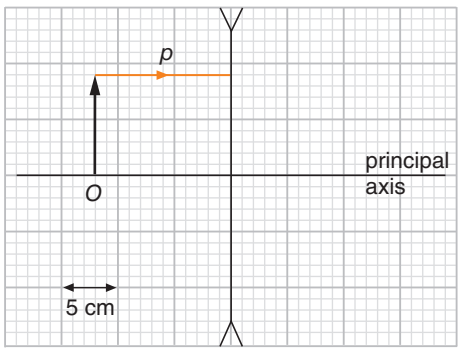


Fig c

(a) Find the image distance.

(b) Draw the image in Figure c.

(c) Complete ray *p*. Hence locate the foci of the lens.

(d) The object is moved towards the lens so that the object distance is smaller than the focal length.

(i) How will the nature of the image change?

(ii) How will the magnification of the image change?

3.4 Relationship between object, image and lens

1 Methods with ray diagram

DSE exam

12(1B)Q7(a), 13(1A)Q22, 13(1B)Q8, 16(1A)Q22, 16(1B)Q5, 18(1B)Q6, 23(1A)Q21

You have now learned the ray diagrams of the two types of lenses and the properties of the images that they form. This allows you to distinguish the two types of lenses and solve more advanced problems.

Example 8 Locating foci and images from one light ray

In Figure a, a light ray s is refracted by a lens L of focal length f. X is a point object lying on the principal axis of the lens.

(a) What type of lens is L? Explain.

(b) Find f.

(c) Where is the image of X?

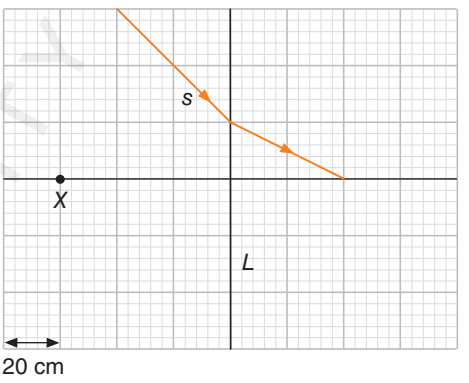


Fig a

Solution

(a) Since ray s is refracted away from the principle axis, L is a concave lens.

(b) As shown in Figure b, f = 40 cm

(c) Suppose an object AB is placed at X (Fig c).

$\therefore$ The image of X is 24 cm from L (at the position of B').

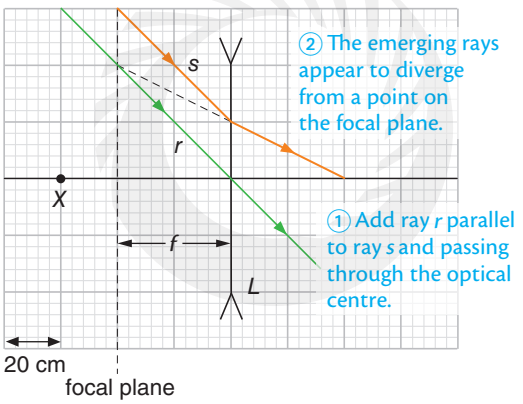


Fig b

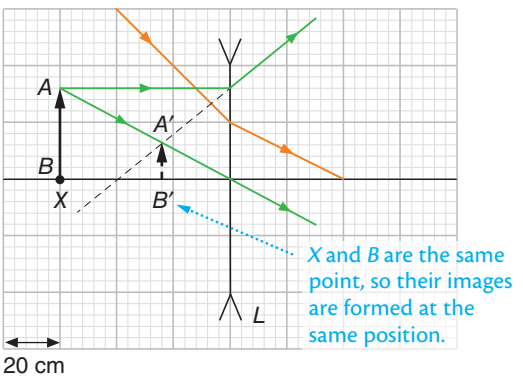


Fig c

OXFORD UNIVERSITY PRESS

Video

→ This video demonstrates a coin box which consists of mirrors and a lens.

Example 9 Wide-angle adapter

A wide-angle adapter of a smartphone is shown in Figure a. Mary studies the adapter and finds that it consists of a lens L. Then she takes photos of the surrounding buildings with (Fig b) and without the adapter (Fig c).



Fig a



Fig b



Fig c

(a) Which type of lens is used in the adapter? Explain your answer.

(b) Mary looks through the adapter directly. In Figure d, AB represents an object in front of the lens L in the adaptor. Light ray p comes from point A and enter Mary's eye after passing through L.

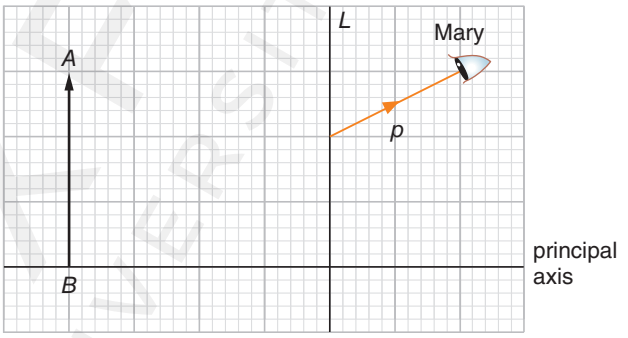


Fig d

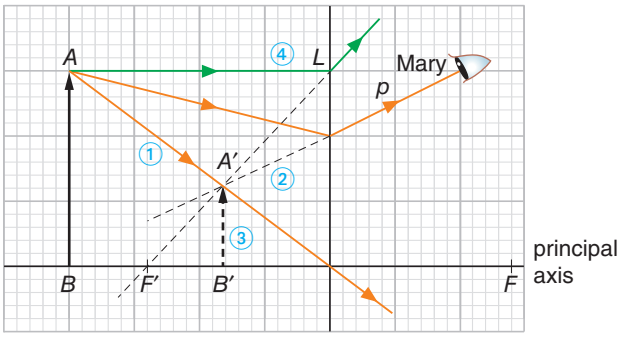
(i) Locate the image of AB. Label it $A'B'$.

(ii) Locate the foci of the lens.

Solution

(a) Since the image is erect and diminished, the lens is a concave lens.

(b) (i), (ii)



Tip

① Draw a ray from A using rule 3.

② Extended ray p backwards.

③ The intersection is A'. Then draw $A'B'$ using a dotted arrow.

④ To locate F', draw a ray from A using rule 1. The emerging ray appears to come from A' and cut the principal axis at F'.

Right and wrong

Video

What will happen to the image formed on the screen if part of the lens is blocked (Fig a)?

✗ Part of the image cannot be seen.

✓ The whole image can be seen but it becomes dimmer (Fig b).

→ This video demonstrates how the brightness of the image changes if part of the lens is blocked.

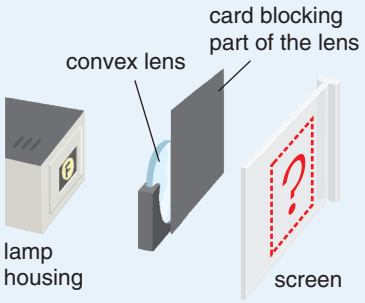


Fig a

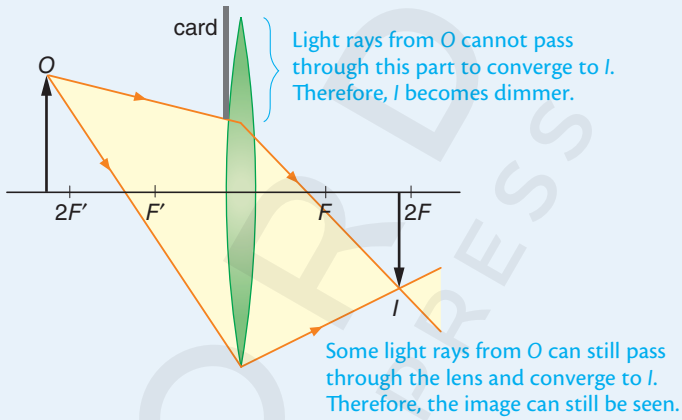


Fig b

→ This is a Video Quiz about the experiment 'Brightness of images formed by a convex lens'.

Learn by practice 5

★1 Point O is the optical centre of the lens (Fig a). Complete rays.

[Hint: To locate F, draw a light ray parallel to q and passing through O.]

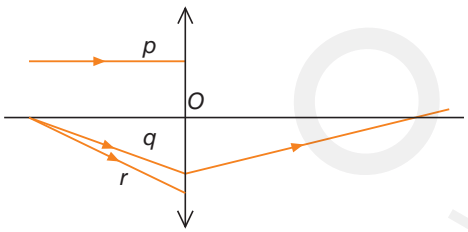


Fig a

★2 Figure b shows object O and its image I formed by a convex lens.



Fig b

(i) Draw a light ray to locate the lens.

(ii) Draw a light ray to locate a focus.

★3 Locate the image of object X (Fig c).

[Hint: Draw an object at X and locate its image.]

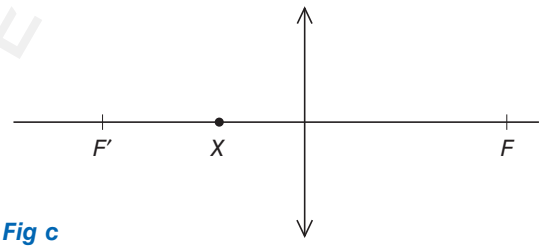


Fig c

★4 An object AB is 10 cm from a concave lens (Fig d). The magnification of the image is 0.4.

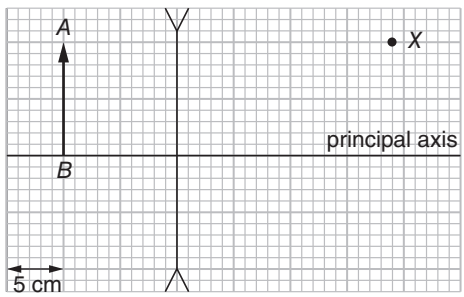


Fig d

(a) Draw light rays to show how the image is formed.

(b) Draw a light ray from point A to point X.

☆ HKDSE 2020 Paper 1A Q20

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Object distance and image distance

An object O is 50 cm away from a lens L (Fig a). A sharp image is formed on a screen S 30 cm from the lens.

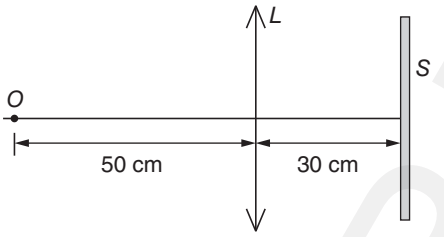


Fig a

Which of the following statements is/are correct?

(1) The focal length of L is shorter than 30 cm.

(2) While fixing S and L, moving O away from L to a certain position can result in a sharp image on S.

(3) While fixing O and S, moving L 20 cm to the left can result in a sharp image on S.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

Solution

A certain image distance only corresponds to a unique object distance.

Common mistake ⚠
Students may not notice this property.

∴ (2) is incorrect.

Figure b shows how an image is formed when O is 50 cm from L.

By the reversibility of light, if O is 30 cm from L, a real image will be formed at a distance 50 cm from L (Fig c).

Common mistake ⚠
Students may overlook that the path of light is reversible.

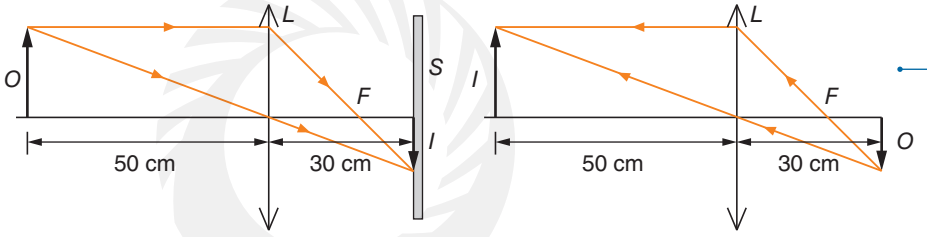


Fig b *Fig c*

When L is moved 20 cm to the left, it is 30 cm from O.

∴ (3) is correct.

From Figure b, the focal length is shorter than 30 cm.

For a real image, both u and v must be larger than f.

∴ (1) is correct.

∴ The answer is B.

▶ Revision 3 Q18 (p.126)

Ext

DSE goal

Apply lens formula to solve problems for a single thin lens (using convention 'real is positive').

DSE exam

12(1B)Q7(b), 15(1A)Q16, 17(1A)Q19, 18(1A)Q19

Simulation

→ This simulation demonstrates the lens formula.

✎ A converging lens is called a positive lens while a diverging lens is called a negative lens.

✎ This formula is also called thin lens formula. For lenses that are not thin, the formula can still be used, but u and v are interpreted differently.

2 Lens formula

→ Start Oxford Physics XR 'Lens' and scan this page to play.

Android iOS

In Chapters 3.2 and 3.3, we learned how to determine the value of u, v or f graphically. In fact, this can be done algebraically using the **lens formula**.

object distance (cm), image distance (cm)

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

focal length (cm)

The lens formula applies to both convex and concave lenses, and it works for both real and virtual images. We follow the **'real-is-positive' convention** when applying the formula (Table 3.4a).

	Convex lens	Concave lens
Object distance u	+	
Image distance v	+ (real image) – (virtual image)	– (virtual image)
Focal length f	+	–

Table 3.4a *The 'real-is-positive' convention for convex and concave lenses.*

Example 10 Lens formula

An object is 15 cm from a convex lens of focal length 10 cm.

(a) Find the image distance. Is the image real or virtual?

(b) If the convex lens is replaced by a concave lens of the same focal length, find the image distance. Is the image real or virtual?

Solution

(a) By $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$,

$$\frac{1}{15} + \frac{1}{v} = \frac{1}{10}$$

$$v = 30 \text{ cm}$$

The image is real since v is positive.

This agrees with the result in Example 4 (p.99).

(b) By $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$,

$$\frac{1}{15} + \frac{1}{v} = \frac{1}{-10}$$

$$v = -6 \text{ cm}$$

The image is virtual since v is negative.

Tip

f is negative for a concave lens.

lens formula 透鏡公式

Ext

Quick check 10

E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 Which sign should be used for different quantities when we apply the lens formula?

Quantity	Sign (+/–)
Object distance	
Image distance of a real image	
Image distance of a virtual image	
Focal length of a convex lens	
Focal length of a concave lens	

2 (a) Complete the table below.

Lens	u / cm	v / cm	f / cm
X	40	10	
Y	3		–6
Z	3	–12	

(b) Are the lenses convex or concave?

(c) Are the images real or virtual?

Learn by practice 6

1 An object is 6 cm from a convex lens of focal length 10 cm. Find the image distance. Is the image real or virtual?

2 An object is 12 cm from a lens. The virtual image formed is 6 cm from the lens. Find the type and the focal length of the lens.

3 An object is 12 cm from a lens. The virtual image formed is 18 cm from the lens. Find the type and the focal length of the lens.

4 An object is in front of a lens L of focal length 20 cm. The image formed is 30 cm from L and is at the same side as the object. What type of lens is L? What is the object distance?

This experiment can deal with real images only.

▶ We can do Experiment 3e to verify the lens formula.

Right and wrong

An object is placed in front of a convex lens and a real image is formed. If the lens is replaced by another convex lens of the same size but with a shorter focal length, how will the brightness of the image change?

✗ It will not change.

✓ The same amount of light passes through the lens (same lens size) but the image formed becomes smaller (shorter focal length). Since the same amount of light is now distributed over a smaller image, the image will be brighter.

Experiment 3e Lens formula

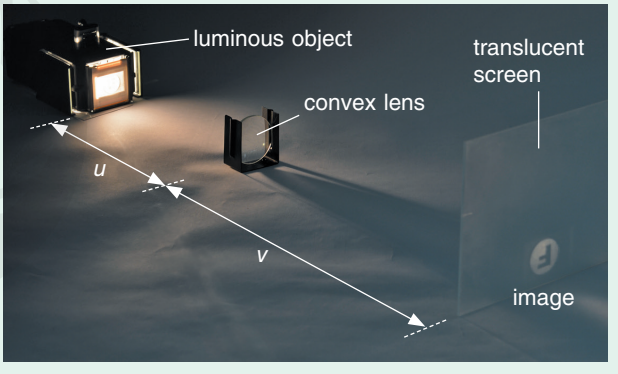
Video

→ This video demonstrates Expt 3e.

1 Use a convex lens of a known focal length f.

2 Put a luminous object in front of the lens and capture the image on a screen. Measure the object distance u and image distance v.

3 Repeat step 2 with different values of u.



Do not touch the hot ray-box lamp.

Discussion

Do the data agree with the lens formula?

→ This is a Video Quiz about this experiment.

Ext

Revision 2 Q33 (p.130) involves another graph that can be obtained in this experiment.

▶ Two graphs can be plotted with the data obtained in Experiment 3e.

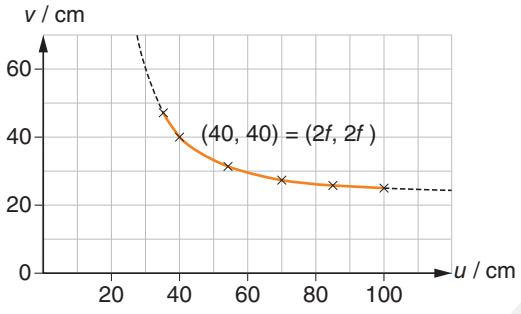


Fig 3.4a Graph of v against u (f = 20 cm).

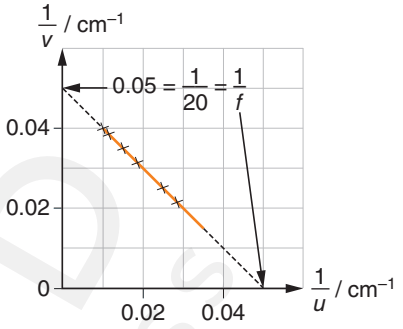


Fig 3.4b Graph of $\frac{1}{v}$ against $\frac{1}{u}$ (f = 20 cm).

1 The graph of v against u is a curve passing though $(2f, 2f)$ (Fig 3.4a).

▶ **2** The graph of $\frac{1}{v}$ against $\frac{1}{u}$ is a straight line of slope –1 (Fig 3.4b). If the graph is extrapolated to the two axes, the intercepts are equal to $\frac{1}{f}$.

3 A certain value of u corresponds to a unique value of v, and vice versa.

By $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$,

$\frac{1}{v} = -\frac{1}{u} + \frac{1}{f}$

$\therefore$ slope = –1

When $\frac{1}{u} = 0$, $\frac{1}{v} = \frac{1}{f}$

When $\frac{1}{v} = 0$, $\frac{1}{u} = \frac{1}{f}$

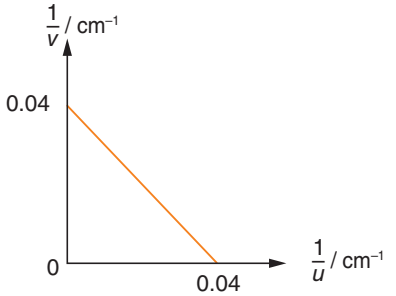
Skill *Mathematics*

Linear relationship

▶ p.137

Example 12 | Graph of $\frac{1}{v}$ against $\frac{1}{u}$

Sam does Experiment 3e and obtains the graph on the right.



(a) Find the focal length of the lens.

(b) Sketch the graph obtained if he repeats the experiment with a lens of a shorter focal length.

Solution

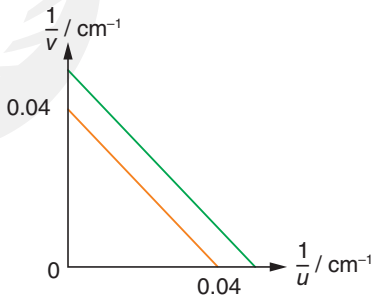
(a) $\frac{1}{f} = 0.04$

$f = 25$ cm

The slope is always –1, so the two lines are parallel.

$f \downarrow \Rightarrow \frac{1}{f} \uparrow \Rightarrow$ intercepts $\uparrow$

(b)



▶ Exercise 3.4 Q2 (p.118)

Ext

☆ HKDSE 2017 Paper 1A Q19

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Convex lens and concave lens

An object is 12 cm in front of a convex lens. The image formed is erect and is 36 cm from the lens.

If the convex lens is replaced by a concave lens of the same focal length, what is the new image distance?

A 5.14 cm B 7.2 cm

C 18 cm D 36 cm

Solution

For the convex lens, by $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$,

$\frac{1}{12} + \frac{1}{(-36)} = \frac{1}{f}$

An erect image must be virtual, so v is negative.

$f = 18$ cm

For the concave lens,

$\frac{1}{12} + \frac{1}{v} = \frac{1}{(-18)}$

Common mistake ⚠
Students may make mistakes in the signs of v and f.

$v = -7.2$ cm

∴ The answer is B.

▶ Revision 3 Q16 (p.125)

Exercise 3.4

★1 A small object X is placed in front of a lens L (Fig a).

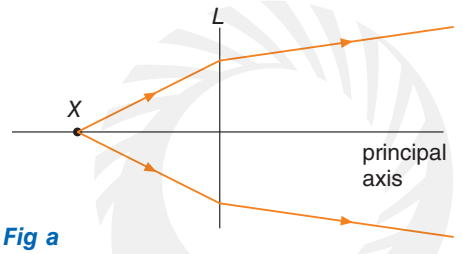


Fig a

Which of the following statements are correct?

(1) L is a concave lens.

(2) The distance of X from the lens is shorter than the focal length.

(3) The image of X lies on the principal axis.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★2 Ext Figure b shows the graph of $\frac{1}{v}$ against $\frac{1}{u}$ of a lens.

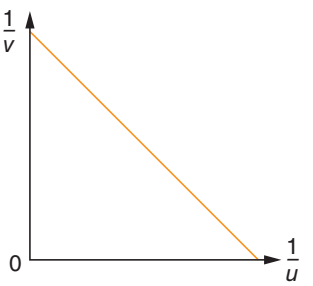


Fig b

Which of the following give(s) $\frac{1}{f}$, where f is the focal length of lens?

(1) The slope of the graph

(2) The $\frac{1}{v}$ intercept of the graph

(3) The $\frac{1}{u}$ intercept of the graph

A (2) only B (3) only

C (2) and (3) only D (1), (2) and (3)

★3

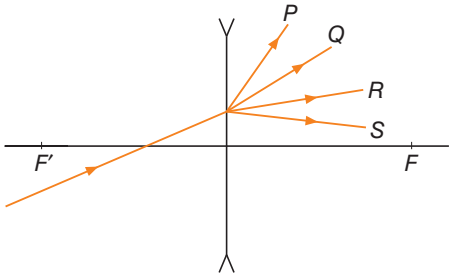


Fig c

A light ray hits a concave lens (Fig c). Which light ray best represents the emerging ray?

A *P*

B *Q*

C *R*

D *S*

★4 Ext An image is formed by a lens as shown (Fig d).

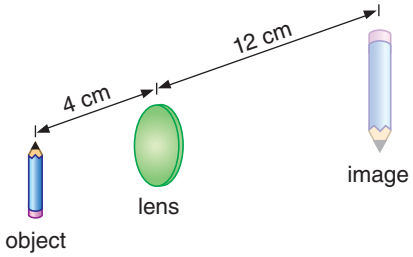


Fig d

What is the focal length of the lens?

A 3 cm

B 4 cm

C 6 cm

D 12 cm

★5 Object *AB* is placed in front of a convex lens (Fig e).

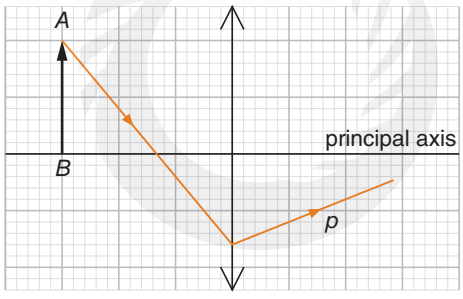


Fig e

(a) Locate the image of *A* by adding a ray.

(b) Locate the foci of the lens.

★6 Ext Figure f shows the *v*–*u* graph of two lenses *P* and *Q*.

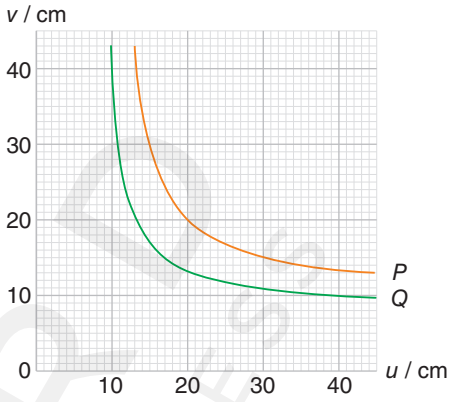


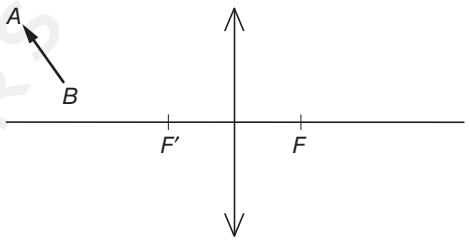
Fig f

(a) Which lens has a longer focal length?

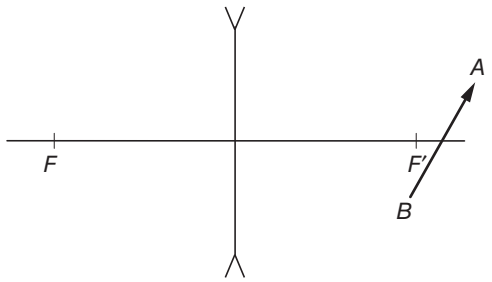
(b) What is the image distance if an object is placed 60 cm from lens *P*?

★7 In each case below, draw light rays to locate the images of points *A* and *B*, hence the image of the whole object.

(a)



(b)



★8 An image formed by a camera wide angle lens is found to be two-thirds the size of the object (head of the man) (Fig g). The object distance is 45 cm.

(a) What kind of lens is the wide-angle lens?

(b) Find the linear magnification of the image.

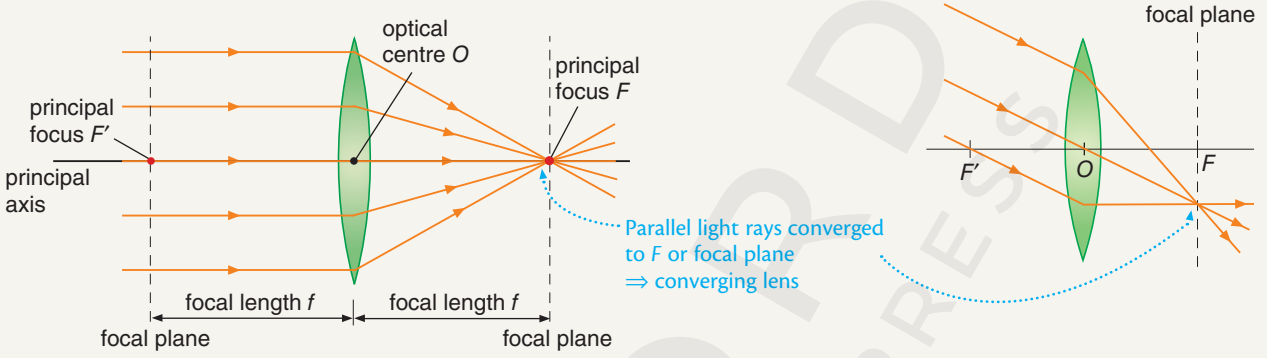
(c) Ext What is the focal length of the lens?



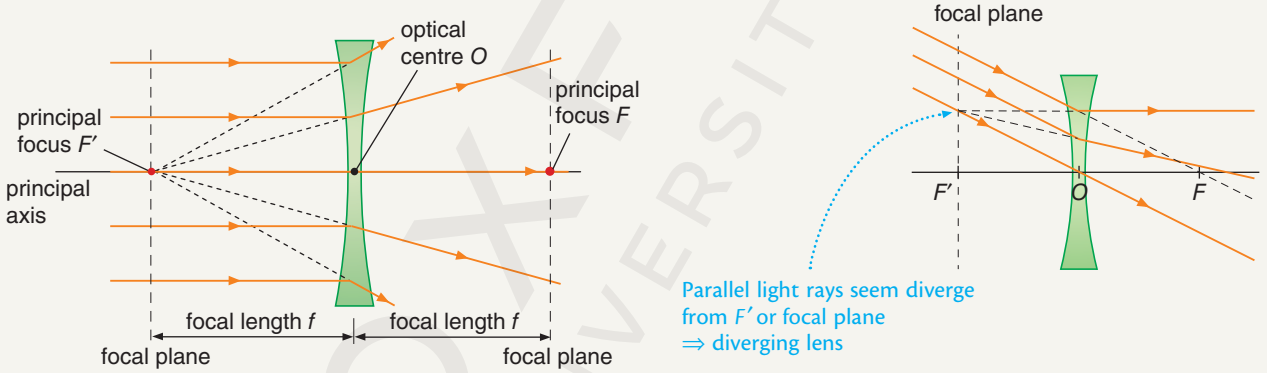
Fig g

Key concepts 3

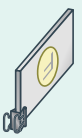
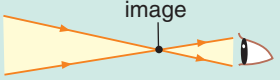
1 Convex lens



2 Concave lens

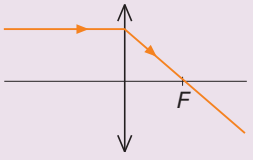
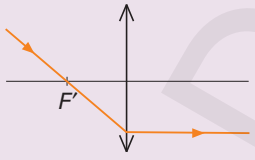
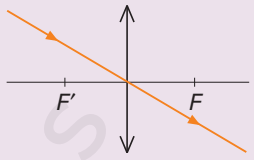
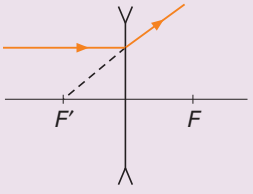
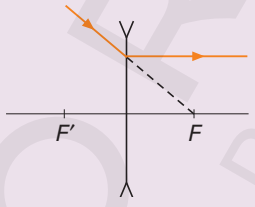
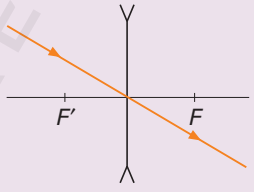


3 Real and virtual images

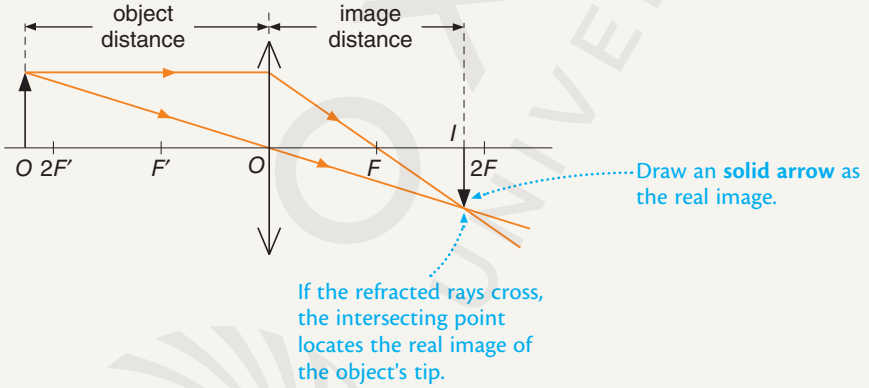
Real image	vs	Virtual image
✓	Can it be projected on a screen? 	✗
✓	Do light rays pass through the image? 	✗
convex lenses only	Formed by	convex lenses, concave lenses, plane mirrors, refraction by water or glass, etc.

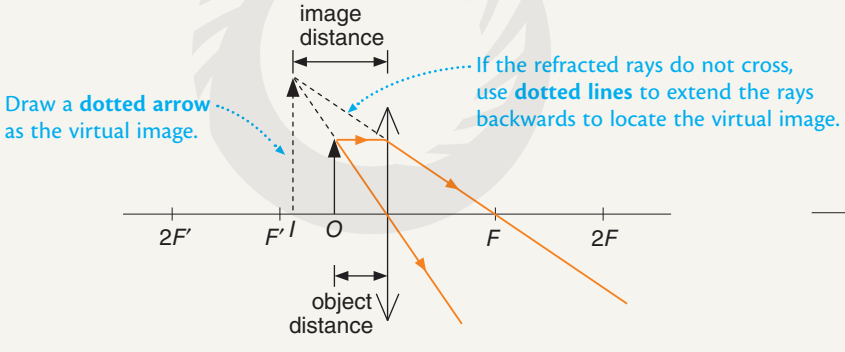
OXFORD UNIVERSITY PRESS

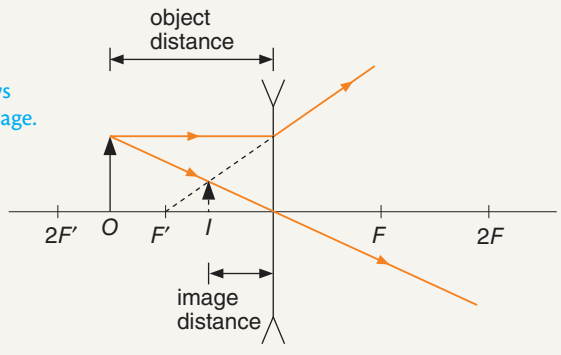
4 Construction rules

	Rule 1	Rule 2	Rule 3
Convex lens			
Concave lens			

5 Graphical construction of images







6 Nature of images

Convex lens	(vs)	Concave lens
Can be real or virtual	**Real or virtual**	Always virtual
Real image: always inverted Virtual image: always erect	**Orientation**	Always erect
Can be same size, magnified or diminished	**Magnification**	Always diminished

7 Linear magnification

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

m	Image size
< 1	diminished
$= 1$	same size as object
> 1	magnified

8 Lens formula Ext

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

	Focal length f	Object distance u	Image distance v	
			Real image	Virtual image
sign	+ (convex lens) − (concave lens)	+	+	−

Follow the 'real-is-positive' convention.

Key terms

1 concave lens 凹透鏡 p.81
2 convex lens 凸透鏡 p.81
4 focal length 焦距 p.83
5 focal plane 焦平面 p.83
6 lens 透鏡 p.80
Ext 7 lens formula 透鏡公式 p.115
8 linear magnification 線性放大率 p.98
10 optical centre 光心 p.83
11 principal axis 主軸 p.83
12 principal focus 主焦點 p.83
13 real image 實像 p.87

Revision 3

Multiple-choice questions

1 Misc Which of the following optical instruments can form virtual images?

(1) Concave lens

(2) Convex lens

(3) Plane mirror

A (1) only

B (2) only

C (1) and (3) only

D (1), (2) and (3)

***2** Misc Lens P and Q form images of Lion Rock which is very far away (Fig a).

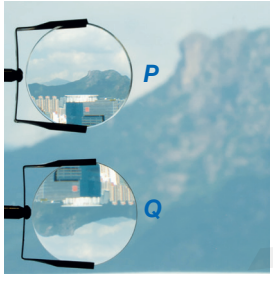


Fig a

Which of the following statements is/are correct?

(1) P is a concave lens while Q is a convex lens.

(2) Both images are virtual.

(3) Both images are formed on the focal planes of the lenses.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

***3** 3.2 A convex lens is used to form an image of an object on a screen. What happens to the image if the upper half of the lens is covered by a piece of cardboard?

A Only the upper half of the image can be seen.

B Only the lower half of the image can be seen.

C The whole image can be seen, but the image becomes dimmer.

D The whole image can be seen, but the image becomes blurred.

***4** 3.2 A student uses the set-up as shown in Figure b to measure the focal length of the convex lens. A sharp image with the same size as the object is captured on the translucent screen.

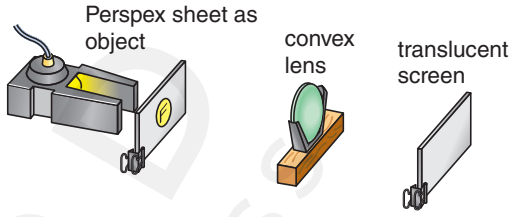


Fig b

Which of the following statements is/are correct?

(1) The captured image is inverted.

(2) The focal length is half the distance between the translucent screen and the lens.

(3) The object is located on the focal plane of the lens.

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

***5** Ext Misc An inverted image is formed by a convex lens. The linear magnification of the image is 5. What is the focal length of the lens in terms of the image distance v?

A $\frac{v}{6}$

B $\frac{v}{4}$

C $\frac{5}{6}v$

D It cannot be determined as the object distance is unknown.

***6** Misc A lens forms an erect image of a distant object. Which of the following statements are correct?

(1) The image is virtual.

(2) The image distance is equal to the focal length of the lens.

(3) The linear magnification of the image is smaller than 1.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★7 3.2 Which of the following statements about the images formed by a convex lens is/are correct?

(1) If the image is virtual, the magnification must be larger than 1.

(2) The image of a distant object must be formed at the focus.

(3) No image is formed when the object is located at $2F$.

A (1) only

B (2) only

C (1) and (2) only

D (1) and (3) only

★8 3.4 Figure c shows two rays, p and q, coming from an object O passing through a lens L. Ray p is parallel to the principal axis and ray q passes through the optical centre of the lens.

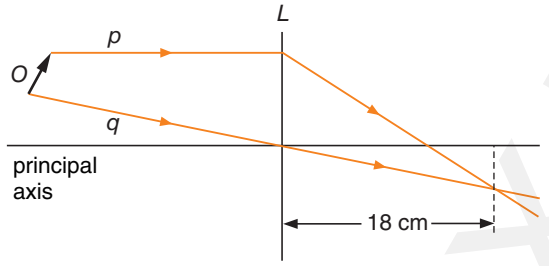


Fig c

Which of the following statements is/are correct?

(1) L is a convex lens.

(2) The image is located at 18 cm from the lens.

(3) The focal length of the lens is shorter than 18 cm.

A (1) only

B (3) only

C (1) and (3) only

D (1), (2) and (3)

★9 3.2 I is an image formed by a convex lens. If the linear magnification of I is greater than 1, which of the following statements must be correct?

(1) I is a real image.

(2) I is inverted.

(3) The image distance is larger the focal length of the lens.

A (1) only B (3) only

C (1) and (2) only D (1), (2) and (3)

★10 Ext 3.4 An object is placed at a distance u from a lens, forming an image at a distance v from the lens on the other side. Which of the following descriptions about the lens is correct?

	Type	Focal length
A	convex	$\frac{uv}{u+v}$
B	convex	$\frac{uv}{u-v}$
C	concave	$\frac{uv}{u+v}$
D	concave	$\frac{uv}{u-v}$

★★11 3.2 A student arranges the apparatus as shown (Fig d). A plane mirror is put behind a convex lens. A sharp image I of the object O (letter F) is formed beside the object. The ray diagram is shown in Figure e.

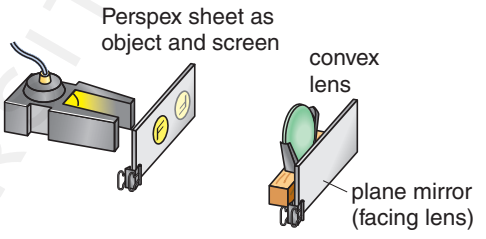


Fig d

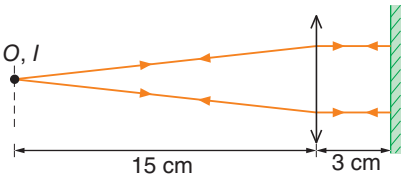


Fig e

Which of the following statements is/are correct?

(1) The focal length of the lens is 15 cm.

(2) O and I are of the same size.

(3) If the distance between the lens and the plane mirror is changed to 15 cm, the position of the image formed remains unchanged.

A (3) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

Refer p.132

**12 Misc

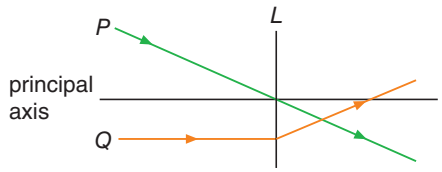


Fig f

Figure f shows two light rays, P and Q, of different colours passing through a lens L. Which of the following statements must be correct?

(1) L is a convex lens.

(2) P does not change its speed when it enters L.

(3) If P and Q are parallel to the principal axis before passing through L, they will converge to the same point on the principal axis.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Refer p.51, 52

**13 3.2 A convex lens is placed in front of an object and an image is formed. The object distance is larger than the focal length. If another convex lens of the same size but with a shorter focal length is used, which of the following statements describing the situation is/are correct?

(1) The image is real.

(2) The image distance decreases.

(3) The image becomes dimmer.

A (1) only

B (2) only

C (1) and (2) only

D (1) and (3) only

Refer p.118

**14 Misc Rays Y and Z passes through a lens as shown (Fig g). Which point will ray X pass through?

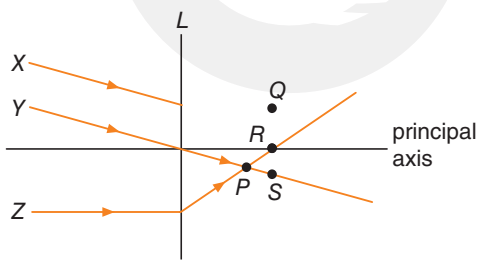


Fig g

A P

B Q

C R

D S

Refer p.94

15 Misc HKDSE 2013 Paper 1A Q22

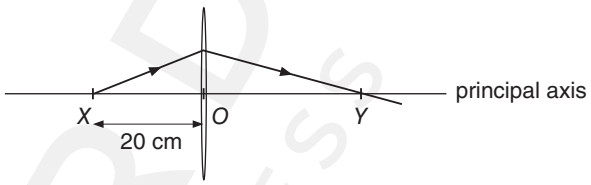


Fig h

A point light source at X on the principal axis of a thin convex lens emits a ray of light. The ray passes through the lens and reaches the principal axis at point Y as shown (Fig h). O is the optical centre of the lens such that OX = 20 cm and $OY > OX$. Which of the following statements is/are correct?

(1) The focal length of the lens is shorter than 20 cm.

(2) If the point light source is shifted away from the lens, separation OY would increase.

(3) An object placed at Y would give a diminished image at X.

A (1) only

B (2) only

C (1) and (3) only (40%)

D (2) and (3) only

16 Misc HKDSE 2017 Paper 1A Q19

When an object is placed 30 cm in front of a concave lens, an image is formed 20 cm away from the lens. If the concave lens is replaced by a convex lens of the same focal length and the object distance remains unchanged, which of the following descriptions about the image formed is correct?

	Nature of image	Image distance
A	real	20 cm
B	real	60 cm
C	virtual	20 cm
D	virtual	60 cm (51%)

17 **HKDSE 2020 Paper 1A Q18**
Misc

Exam report: Candidates choosing options B and D suggests that they did not have a basic understanding of the converging and diverging properties of lenses.

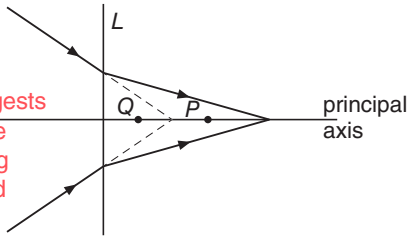


Fig i

Referring to the above ray diagram (Fig i), what kind of lens is represented by L? Which point, P or Q, can be its focus?

	lens L	focus
A	concave	P (45%)
B	convex	P
C	concave	Q
D	convex	Q

18 **HKDSE 2020 Paper 1A Q20**
3.3

Figure j shows an enlarged sharp image of an object O formed on a screen S by a convex lens L.

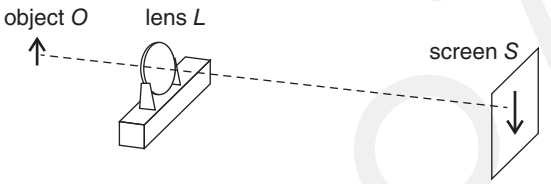


Fig j

Which of the following can give a diminished sharp image on the screen?

(1) Keeping the positions of O and L unchanged, move S suitably closer to L.

(2) Keeping the positions of L and S unchanged, move O suitably farther away from L.

(3) Keeping the positions of O and S unchanged, move L suitably closer to S.

A (1) only

B (3) only (31%)

C (1) and (2) only

D (2) and (3) only

Exam report: Less than one-third of the candidates were able to answer this question correctly. It seems that many candidates were not aware that a certain object distance only corresponds to a unique image distance.

Conventional questions

★**19** Figure k shows the positions of an object O and its image I formed by a convex lens.
3.2

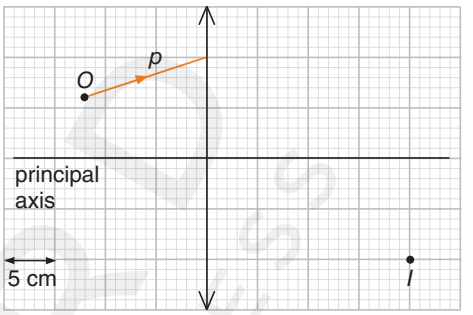


Fig k

(a) Draw a light ray to find the focal length of the lens. (2 marks)

(b) State the nature of the image. (2 marks)

(c) Complete ray p. (1 mark)

(d) If the object is moved 7 cm towards the lens, is the image formed by the lens virtual or real? Explain your answer. (2 marks)

★**20** An object is 40 cm from a convex lens, whose focal length is 10 cm.
Ext
3.4

(a) Find the image distance. (2 marks)

(b) Then the object moves towards the lens with a constant speed.

(i) Find the image distance when the object is 35 cm from the lens. (1 mark)

(ii) Does the image move towards or away from the lens? (1 mark)

(iii) Does the image move faster or slower than the object when the object distance is 35 cm? (1 mark)

(iv) When the object distance changes from 15 cm to 14 cm, how much does the image distance change? Does the image move faster or slower than the object? (2 marks)

(c) Then the object moves towards a concave lens with a constant speed.

(i) To which direction does the image move? (1 mark)

(ii) Does the image move faster or slower than the object? (1 mark)

★21 3.4 An object O is placed in front of a lens L. Its image I is formed as shown (Fig l).

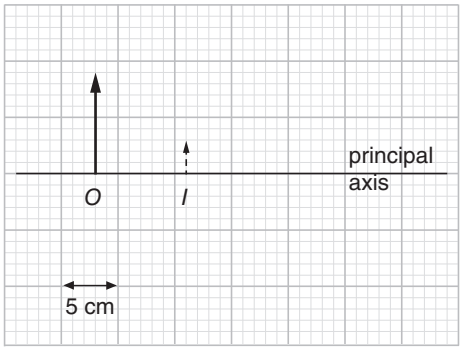


Fig l

(a) What kind of lens is X? Explain. (2 marks)

(b) Draw a ray to locate L. (1 mark)

(c) Draw a ray to locate the focus and hence, find the focal length of the lens. (2 marks)

★22 Misc An object O is placed in front of a lens L. An enlarged image I is formed on the same side as the object (Fig m).

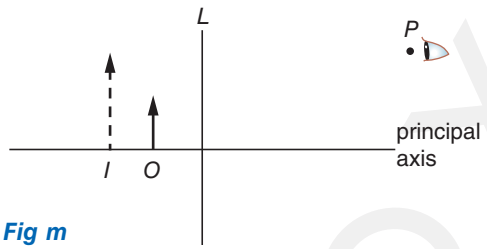


Fig m

(a) What kind of lens is L? Explain. (2 marks)

(b) Draw two rays to show how an observer at point P sees the whole object. (2 marks)

(c) If the object is moved a little closer towards L, how will the image change? Support your answer by drawing a ray diagram. (3 marks)

★23 Ext 3.4 A student put an object in front of a convex lens and use a screen to capture the image formed. She varies the object distance u and measures the corresponding image distance v.

(a) Sketch a graph of $(u + v)$ against uv. (3 marks)

(b) How can you find out the focal length of the lens from the graph in (a)? (1 mark)

(c) Can this method be used to find out the focal length of a concave lens? Explain. (2 marks)

★24 Misc In Figure n, AB represents an object placed in front of a diverging lens. P, Q, R and S are incident rays from AB to the lens. XX' is the principal axis of the lens. When AB is 32 cm from the lens, its image is formed 15 cm from the lens.

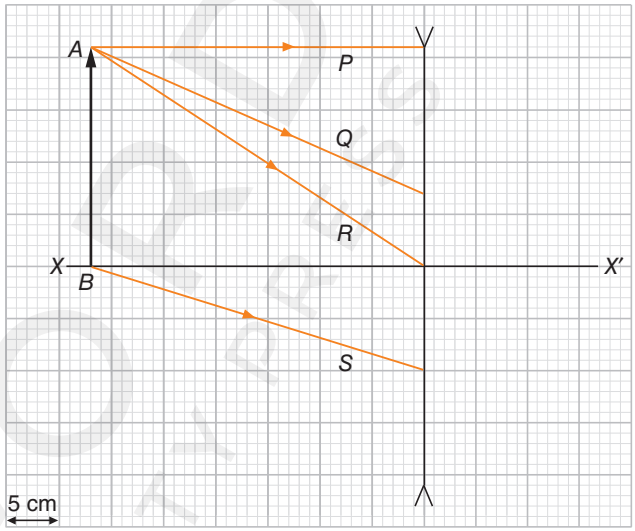


Fig n

(a) State the nature of the image. (3 marks)

(b) Find the linear magnification. (1 mark)

(c) Draw the image of AB and the refracted rays for P, Q, R and S in the figure. (2 marks)

(d) Find the focal length of the lens. (1 mark)

(e) The diverging lens is then changed to a converging lens with the same focal length.

(i) Ext Use the lens formula to find the image distance. (2 marks)

(ii) State the nature of the image. (3 marks)

★25 3.4 Four rays p, q, r and s from an object O incident on lens L as shown (Fig o).

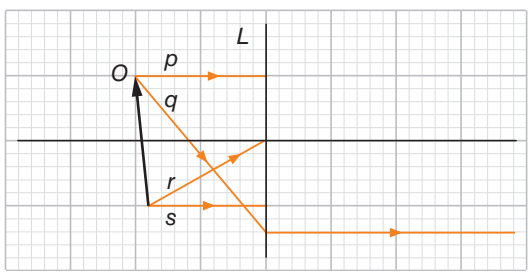


Fig o

(a) Mark and label the two foci, F and F', of L in Figure o. (2 marks)

(b) Complete the rays p, r and s. (3 marks)

(c) Locate the image of object O. (1 mark)

★26 Misc Read the following passage about the **lens of an eye** and answer the questions that follow.

The lens is a transparent, convex structure in the eye. Along with the cornea, it helps refract light to be focused on the retina. By changing its shape, the lens allows us to focus on objects at various distances. To see a nearby object, the ciliary muscles contract and the lens becomes thick (Fig p). Changing focus to better see another object further away requires the stretching of the lens by the relaxation of the ciliary muscles. The lens then becomes thinner and the focal length increases (Fig q).

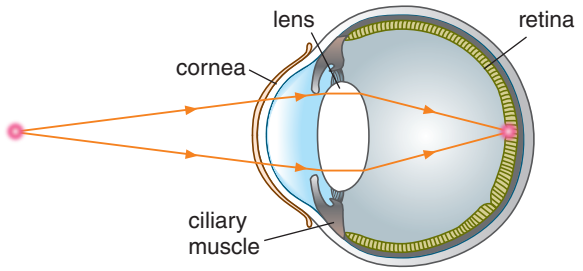


Fig p

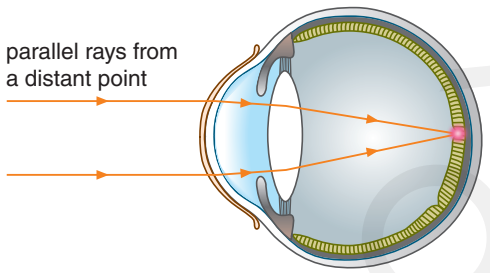


Fig q

(a) What problem would occur if our lenses were concave? (1 mark)

(b) Suppose the lens is 2.5 cm from the retina. For simplicity, ignore the refraction by the cornea.

(i) Ext When looking at an object 30 cm in front of the eye, what is the focal length of the lens? (2 marks)

(ii) When the eye looks at another object which is 1 m away, how does the eyes perform accommodation? (2 marks)

★27 Misc In a single-lens reflex camera (Fig r) the distance between the lens and the image sensor can be adjusted so that objects at different locations can be put into focus.

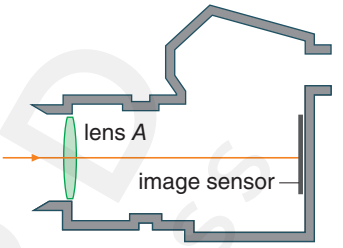


Fig r

An object 1.2 m in front of the camera is to be photographed.

(a) Explain what kind of lens *A* is. (2 marks)

(b) The focal length of lens *A* is 200 mm.

(i) Ext Find the distance between lens *A* and the image sensor by using the lens formula. (2 marks)

(ii) Find the linear magnification of the image. (2 marks)

(c) Now lens *A* is replaced with lens *B*, which is the same type of lens but with a different focal length. The linear magnification of the image becomes larger.

(i) How should the distance between lens *B* and the sensor change so that a clear image of the object can be formed on the sensor? (1 mark)

(ii) In order to get a wider view, should lens *A* or *B* be used? Explain your answer. (2 marks)

★28 Misc Figure s shows a view through a peephole on a front-entrance door. Suppose there is only one lens in the peephole.



Fig s

(a) What kind of lens is used in the peephole? Explain your answer. (2 marks)

(b) State an advantage of using the lens in (a) in the peephole. (1 mark)

(c) Ext Suppose the lens in the peephole has a focal length of 5 cm. When a visitor is 1 m away from the lens, where is the image of the visitor? (2 marks)

★ 29 3.2 An object AB is placed 10 cm away from a convex lens (Fig t). A virtual image is formed and is 20 cm from the lens.

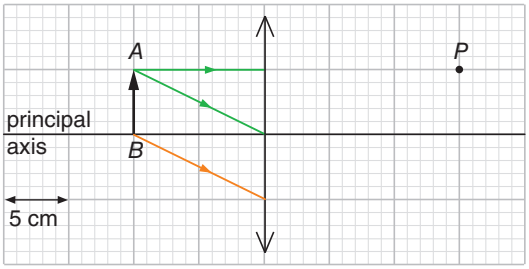


Fig t

(a) Find the linear magnification of the image. (2 marks)

(b) Complete the rays in Figure t. (4 marks)

(c) Find the focal length of the lens. (1 mark)

(d) Draw a ray to show how an observer at point P sees the image of A. (1 mark)

★★ 30 3.2 An object O is placed 36 cm from a convex lens (Fig u). A ray from O is refracted by the lens as shown.

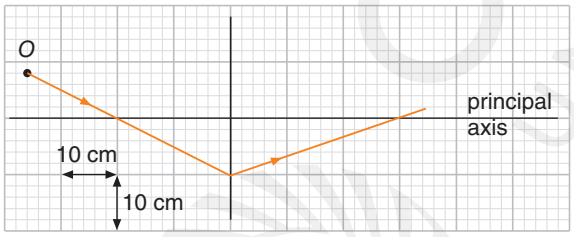


Fig u

(a) Find the focal length of the lens. (2 marks)

(b) Locate the image of O. (2 marks)

(c) Find the linear magnification of the image. (2 marks)

(d) Suggest two changes of the image if

(i) a convex lens with a shorter focal length is used; (2 marks)

(ii) a convex lens with a focal length longer than 36 cm is used; (2 marks)

Refer Example 8 (p.111)

★★ 31 3.3 Figure v shows an object AB placed in front of a small concave lens.

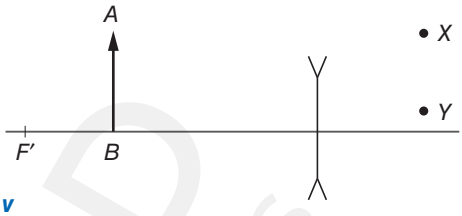


Fig v

(a) Locate the image graphically. (3 marks)

(b) Can observers at points X and Y see the whole image? Illustrate your answer graphically. (3 marks)

Refer p.113

★★ 32 Misc A student captures the image of an illuminated object formed by a lens on a screen using the set-up shown in Figure w. He varies the object distance u and measures the corresponding image distance v. The results are plotted on a graph of v against u (Fig x).

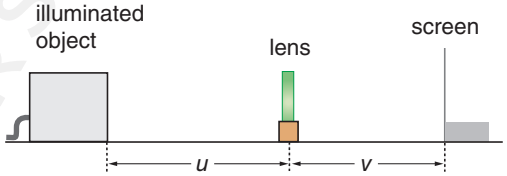


Fig w

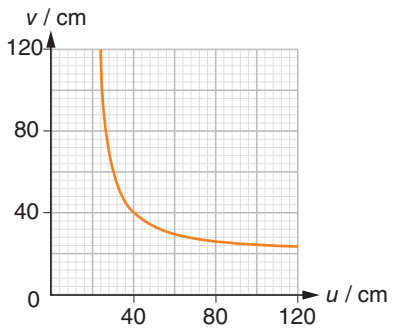


Fig x

(a) How does he ensure that the position of the screen is correct? (1 mark)

(b) What kind of lens did he use? Explain. (2 marks)

(c) What value for the focal length of the lens does the graph suggest? Explain. (2 marks)

(d) Sketch a straight-line graph relating v and u. Mark the values of the intercepts. (4 marks)

Refer p.117

33 HKDSE 2016 Paper 1B Q5

Misc

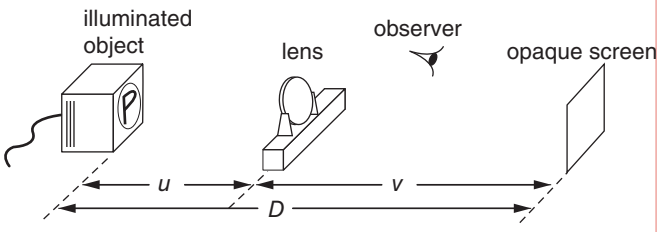


Fig y

Kitty uses the set-up in Figure y to study the image formation of a lens. The lens is placed at a distance u from an illuminated object (letter '*P*'). An opaque screen is placed at a distance D from the object to capture the image.

(a) (i) State the kind of lens used. Explain your answer. (2 marks)

(ii) Sketch the image on the screen seen by the observer. (1 mark)

(b) The separation D is varied while the position of the lens is adjusted to form an image once again on the screen. The corresponding object distance u is obtained for plotting a graph of D against u (Fig z).

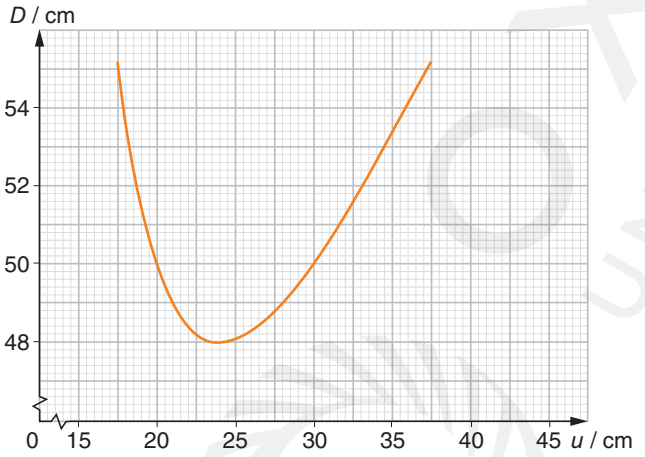


Fig z

(i) When the lens is placed at 18 cm from the object, use the graph to find the corresponding separation between the lens and the screen. Hence calculate the magnification of the image. (2 marks)

In Figure aa, AB represents the illuminated object which is at 18 cm from the lens L. p, q and r are light rays from AB.

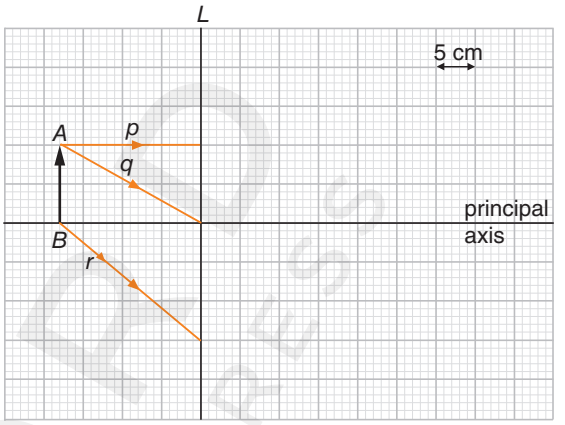


Fig aa

(ii) Indicate the image formed by AB (denote it as I) and draw the refracted rays of p, q and r. (3 marks)

(iii) Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

(iv) Keeping the object and screen fixed in position, suggest where Kitty should move the lens to such that an image can be formed again on the screen. State the ratio $\frac{\text{height of this new image}}{\text{height of the original image}}$. (2 marks)

34 HKDSE 2018 Paper 1B Q6

3.3

(a) You are given a ray box with a single slit (producing a fine light beam), a cylindrical concave lens, a plastic ruler, a pencil and a piece of paper as shown in Figure ab.

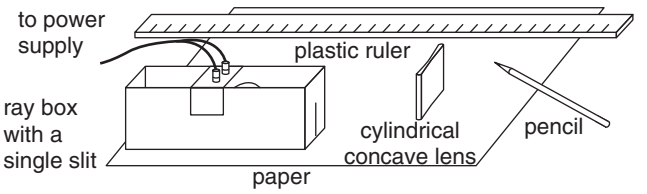


Fig ab

Describe how you would use the above apparatus to find the focal length of the lens and state ONE possible source of error in the experiment. (5 marks)

Q33 Exam report: Candidates performed well in (a)(i). In answering (a)(ii), many candidates merely memorized the image formed by a convex lens without considering the observer's position. Part (b) was in general well answered. Some candidates made mistakes in reading the distance D form the graph. Quite a number of them did not realize that the object and image distances could be interchanged in (b)(iv). Many mistook the magnification of the new image as the height ratio regarding the two situations involved.

(b) In the Figure ac, L represents another cylindrical lens. A vertical pin used as the object is placed at O, the image is formed at I by the lens. The horizontal scale is 1 cm to 10 cm.

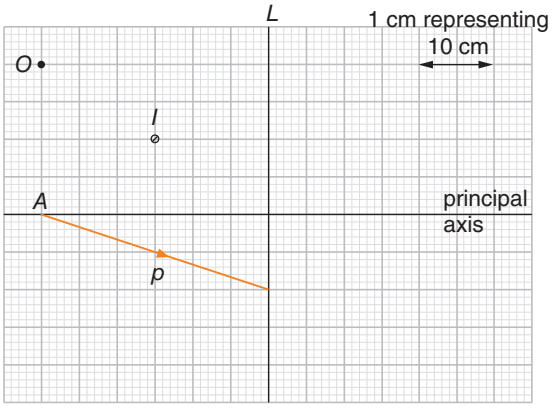


Fig ac

(i) What kind of lens is used? Explain. (2 marks)

(ii) Draw a suitable light ray to locate the principal focus F of lens L. Find its focal length. (2 marks)

(iii) Complete the path for the ray p from point A. (1 mark)

Q34 Exam report: Candidates' performance in (a) was poor. Not many of them correctly suggested various methods, using one ray (with the lens position and principal axis correctly marked on the piece of paper) or two parallel rays from the ray box to find the focal length of the concave lens. Weaker candidates, however, wrongly tried to obtain an image formed by the concave lens using just a single light ray from the ray box. Part (b) was in general well answered. Most were able to draw a suitable light ray to find the focal length of the lens in (b)(ii). In (b)(iii), however, some candidates failed to indicate the correct path of ray p.

35 **HKDSE 2019 Paper 1B Q6**
3.4

In Figure ad, AB represents the virtual image of an object formed by lens L. The magnification of the image is 0.4. The horizontal scale is 1 cm to 10 cm.

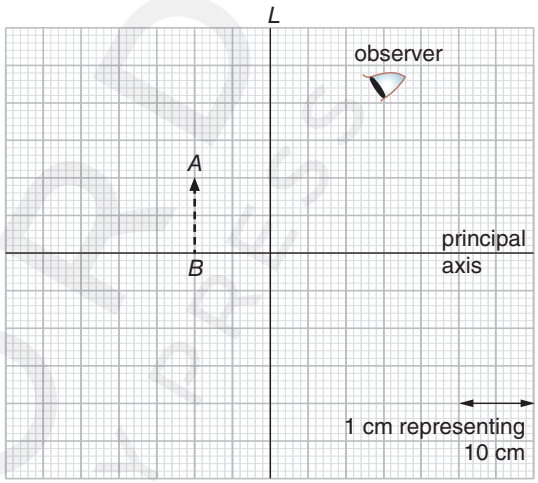


Fig ad

(a) What kind of lens is used? Explain. (2 marks)

(b) Indicate on Figure af the position and height of the object. (2 marks)

(c) By drawing a suitable light ray, locate and mark the position of the focus, F, of the lens. Find the focal length of the lens. (3 marks)

(d) Draw a light ray emerging from the object to illustrate how the observer in the figure can see the tip A of the image. (2 marks)

Q35 Exam report: This question tested candidates' knowledge and understanding on image formation by a concave lens. Candidates' performance was fair. Most realized that the lens was a concave one and gave valid explanation in (a). However, some made mistakes in the position and height of the object in (b) and thus failed to locate the correct position of the focus in (c).

Part (d) was poorly answered. Not many were able to draw a correct light ray from the object so as to illustrate how the image's tip could be observed.

Know more

Measuring *f* (p.94)

The focal length of a convex lens can be found by setting up the apparatus as shown (Fig a). The position of the lens is adjusted so that a sharp image of *X* is formed next to *X*. The distance between the *X* and the lens is the focal length of the lens.

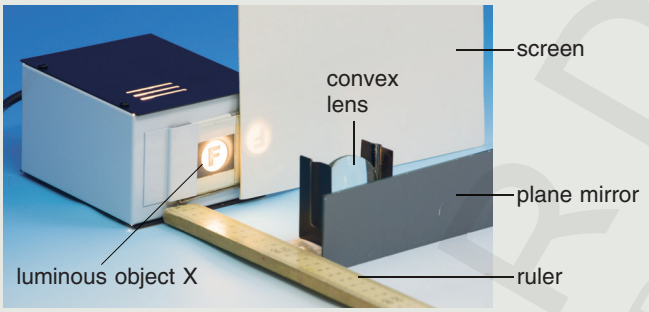


Fig a

The working principle of this method can be explained as follows (Fig b).

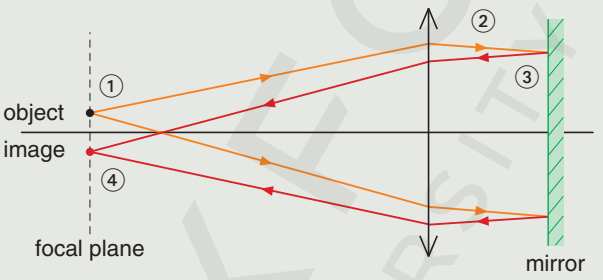


Fig b

① The object is on the focal plane.

② Light rays from the object become parallel after passing through the lens.

③ On hitting a plane mirror behind the lens, the rays are reflected back, still parallel to each other.

④ The reflected parallel rays pass through the lens and form an image on the focal plane next to the object.

Cylindrical and spherical convex lenses (p.97)

A cylindrical convex lens is always used to illustrate the construction rules for a convex lens. However, it is different from a spherical convex lens. A cylindrical lens converges parallel rays in the horizontal plane and forms a vertical line. A spherical lens converges parallel rays in both the horizontal and vertical plane and forms a dot (Fig c).

When a real image is formed by a cylindrical convex lens, it is only laterally inverted (Fig d). On the other hand, a real image formed by a spherical convex lens is inverted and laterally inverted (Fig e).

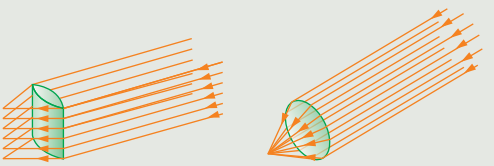


Fig c



Fig d



Fig e